PROFESSOR DIMITRIOS S. NIKOLOPOULOS
John W. Hancock Professor of Engineering, IEEE Fellow
Professor, Department of Computer Science
Professor (by courtesy), Department of Electrical and Computer Engineering
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Employment History

John W. Hancock Professor in Engineering, Virginia Tech, Aug. 2019–present
Professor, Department of Computer Science, Virginia Tech, Aug. 2019–present
Professor (by courtesy), Department of Electrical and Computer Engineering, Virginia Tech, Aug. 2020–present
Associate Director, Stacks@CS Center for Computer Systems Research, Virginia Tech, Aug. 2022–present
Honorary Professor, School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast, Aug. 2022–present
Director, Queen's University Belfast Research Institute in Electronics, Communications and Information Technology (ECIT), Jan. 2018–Aug. 2019
Head of School, School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast, Jan. 2016–Jan. 2018
Royal Society Wolfson Research Fellow, Sep. 2015–present
Professor & Chair in High Performance and Distributed Computing, School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast, Jan. 2012–Aug. 2019
Director, Center for Data Science and Scalable Computing, Queen's University Belfast, Feb. 2016–Aug. 2019
Director, High Performance and Distributed Computing Research Cluster, Queen's University Belfast, Jan. 2012–Feb. 2016
Adjunct Professor, Department of Computer Science, Virginia Tech, Oct. 2016–Aug. 2019
Adjunct Professor, Department of Computer Science, Old Dominion University, Oct. 2013–present
Associate Professor with Tenure, Department of Computer Science, University of Crete, Sep. 2009–Jan. 2012
Affiliate Professor, Institute of Computer Science Foundation for Research and Technology (FORTH), Sep. 2009–Jan. 2012
Associate Professor with Tenure, Department of Computer Science, Virginia Tech, Aug. 2008–Sep. 2009
Associate Professor Tenure-Track, Department of Computer Science, Virginia Tech, Aug. 2006–Aug. 2008
Assistant Professor Tenure-Track, Department of Computer Science, College of William & Mary, Aug. 2002–Aug. 2006
Visiting Assistant Professor, Department of Electrical and Computer Engineering, University of Illinois, Urbana-Champaign, Jan. 2001–Aug. 2002

Earned Degrees

Ph.D.	Computer Engineering and Informatics, University of Patras, 2000
Master's in Engineering	Computer Engineering and Informatics, University of Patras, 1997
Diploma in Engineering[1]	Computer Engineering and Informatics, University of Patras, 1996

[1] A Diploma in Engineering in Greece is a 5-year University degree that is academically equivalent to a qualification above the Bachelor's and below the Master's degree in the US.

Primary Research Areas

High Performance Computing and Computational Science; Computer Systems

Current Technical Research Interests

- **Efficient, resilient AI systems across the computing continuum:** federated learning, edge/microcloud infrastructure, and runtime/orchestration that make AI robust under heterogeneity and failures.
- **Performance engineering for modern accelerators:** GPU/CPU overlap, memory/KV-cache redesign, long-context inference/training efficiency, and cost/energy-aware optimization.
- **Systems for real-world, data-intensive science and cyberinfrastructure:** streaming + learning pipelines, scalable experimentation/digital twins, and deployable research platforms with measurable impact.

Honors and Awards

[43] **Sigma Xi Full Member**, The Scientific Research Honor Society, 2025

[42] **IEEE Fellow**, "For contributions to dynamic execution environments and multiprocessor memory management", 2024

[41] **IEEE Computer Society Distinguished Visitor**, 2024

[40] **Fellow, International Artificial Intelligence Industry Alliance**, 2024

[39] **Fellow, Asia-Pacific Artificial Intelligence Association**, 2023

[38] **Distinguished Contributor, IEEE Computer Society**, 2021

[37] **Best Paper Award**, Design Automation and Test in Europe Conference (DATE), 2020

[36] **IEEE Award for Editorial Excellence, IEEE Transactions on Parallel and Distributed Systems**, 2020

[35] **HiPEAC Paper Award**, twice in 2020 (for ASPLOS'2020 and MICRO'2020 papers)

[34] **Best Paper Award Nomination**, IEEE/ACM International Symposium on Microarchitecture (MICRO), 2020

[33] **John W. Hancock Chair in Engineering**, Virginia Tech, 2019–present

[32] **Elsevier Distinguished Editorial Service Award**, 2019

[31] **Distinguished Member of the ACM**, 2018

[30] **IEEE Outstanding Service Award**, 2018

[29] **Fellow of the IET**, 2017

[28] **Investors in People Silver Award**, 2017

[27] **Royal Society Wolfson Research Merit Award**, 2015 (extended to lifetime)

[26] **SFI-DEL Investigator Award**, 2015

[25] **Fellow of the British Computer Society**, 2014

[24] **IEEE Outstanding Service Award**, 2014

[23] **Best Paper Award**, ACM International Workshop on Code Optimization for Multi and Many Cores (COSMIC), 2013

[22] **ACM Senior Member**, 2011

[21] **IEEE Senior Member**, 2010

[20] **IEEE Outstanding Service Award**, 2010

[19] **Best Paper Award Nomination**, International Conference on Parallel Processing (ICPP), 2010

[18] **Marie Curie Fellow**, 2009

[17] **HiPEAC Fellow**, 2008

[16] **IBM Faculty Award**, 2007

[15] **Best Paper Award**, ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPOPP), 2007

[14] **Best Paper Nomination**, ACM Symposium on High-Performance Parallel and Distributed Computing (HPDC), 2006

[13] **DOE Early Career Principal Investigator Award**, 2005

[12] **Best Paper Award**, International Workshop on OpenMP (IWOMP), 2005

[11] **NSF CAREER Award**, 2004

[10] **Best Paper Award**, International Symposium on High-Performance Computing (ISHPC), 2003

[9] **Best Paper Award**, IEEE/ACM International Symposium on Cluster Computing and the Grid (CCGRID), 2002

[8] **Best Paper Award**, IEEE/ACM International Parallel and Distributed Processing Symposium (IPDPS), 2002

[7] **Best Paper Award Nomination**, ACM International Conference on Supercomputing (ICS), 2001

[6] **Best Paper Award Nomination**, IEEE/ACM Supercomputing: High-Performance Computing, Networking, Storage, and Analysis Conference (SC), 2001

[5] **Best Paper Award**, IEEE/ACM Supercomputing: High-Performance Computing, Networking, Storage and Analysis Conference (SC), 2000

[4] **Best Paper Award Nomination**, ACM International Conference on Supercomputing (ICS), 1999

[3] **Outstanding Academic Performance Award**, Technical Chamber of Greece, 1996

[2] **Outstanding Academic Performance Award**, Technical Chamber of Greece, 1993

[1] **Outstanding Academic Performance Award**, National Scholarships Foundation of Greece, 1992

Publications

Refereed Publications

Journal Articles

[282] X. Li, S. Ghafouri, H. Vandierendonck, D. John, and D. S. Nikolopoulos, "Sweet: Serving workload-balanced end-to-end efficient and tailored edge inference via quantization and partitioning," ***Frontiers in Complex Systems***, Apr. 2026, Section on Multi- and Cross-Disciplinary Complexity.

[281] X. Li, J. Fan, Q. Wang, D. Spatharakis, S. Ghafouri, H. Vandierendonck, D. John, B. Ji, A. R. Butt, and D. S. Nikolopoulos, "Wisp: Waste- and interference-suppressed distributed speculative llm serving at the edge via dynamic drafting and slo-aware batching," ***Proceedings of the ACM on Measurement and Analysis of Computing Systems (ACM POMACS-SIGMETRICS)***, 2026.

[280] A. Kumar, C. O'Mahoney, P. K. Werle, S. Shanker, D. S. Nikolopoulos, B. Ji, H. Vandierendonck, and D. John, "Marvel: An end-to-end framework for generating model-class aware custom risc-v extensions for lightweight ai," ***IEEE Open Journal of Circuits and Systems***, vol. 6, pp. 445–456, 2025.

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[278] E. A. Mazied, D. S. Nikolopoulos, Y. Hanafy, and S. F. Midkiff, “Auto-scaling edge cloud for network slicing,” ***Frontiers in High Performance Computing***, vol. 1, 2023. DOI: https://doi.org/10.3389/fhpcp.2023.1167162

[277] C.-H. Hong, M. Lee, H. Ahn, and D. S. Nikolopoulos, “Gshare: A centralized gpu memory management framework to enable gpu memory sharing for containers,” ***Future Generation Computer Systems***, vol. 130, pp. 181–192, May 2022.

[276] K. Dichev, D. di Sensi, I. Spence, K. W. Cameron, and D. S. Nikolopoulos, “Power log’n’roll: Power-efficient localized rollback for mpi applications using logging protocols,” ***IEEE Transactions on Parallel and Distributed Systems***, vol. 33, no. 6, pp. 1276–1288, Jun. 2022.

[275] U. Minhas, R. Woods, D. S. Nikolopoulos, and G. Karakonstantis, “Efficient, dynamic multi-task execution on fpga-based computing systems,” ***IEEE Transactions on Parallel and Distributed Systems***, vol. 33, no. 3, pp. 710–722, Mar. 2022.

[274] L. Mukhanov, K. Tovletoglou, H. Vandierendonck, D. S. Nikolopoulos, and G. Karakonstantis, “Revealing dram operating guardbands through workload-aware error predictive modeling,” ***IEEE Transactions on Computers***, no. 11, pp. 1976–1987, Nov. 2021.

[273] J. Lee, H. Vandierendonck, and D. S. Nikolopoulos, “Mixed precision kernel recursive least squares,” ***IEEE Transactions on Neural Networks and Learning Systems***, no. 3, pp. 1284–1298, Mar. 2022. DOI: 10.1109/TNNLS.2020.3041677

[272] J. Sun, H. Vandierendonck, and D. S. Nikolopoulos, “Fast load balance parallel graph analytics with an automatic data structure selection algorithm,” ***Future Generation Computer Systems***, vol. 112, pp. 612–623, Nov. 2020.

[271] J. Lee, G. T. Peterson, H. Vandierendonck, and D. S. Nikolopoulos, “Air: Iterative refinement acceleration using arbitrary dynamic precision,” ***Parallel Computing***, vol. 97, Sep. 2020, Article: 102663.

[270] N. Wang, B. Varghese, M. Matthaiou, and D. S. Nikolopoulos, “Dyverse: Dynamic vertical scaling in multi-tenant edge environments,” ***Future Generation Computer Systems***, vol. 108, pp. 598–612, Jul. 2020.

[269] H. Vandierendonck and D. S. Nikolopoulos, “Hyperqueues: Design and Implementation of Deterministic Concurrent Queues,” ***ACM Transactions on Parallel Computing***, vol. 6, no. 4, pp. 1–35, Nov. 2019, Article 23.

[268] A. Hassan, H. Vandierendonck, and D. S. Nikolopoulos, “Fast and Energy-Efficient OLAP Data Management on Hybrid Main Memory Systems,” ***IEEE Transactions on Computers***, vol. 63, no. 11, pp. 1597–1611, Nov. 2019.

[267] I. Tsiokanos, L. Mukhanov, D. S. Nikolopoulos, and G. Karakonstantis, “Significance-Driven Data Truncation for Preventing Timing Failures,” ***IEEE Transactions on Device and Materials Reliability***, vol. 19, no. 1, pp. 25–36, Mar. 2019.

[266] K. Tovletoglou, L. Mukhanov, D. S. Nikolopoulos, and G. Karakonstantis, “Shimmer: Implementing a Heterogeneous-Reliability DRAM Framework on a Commodity Server,” ***IEEE Computer Architecture Letters***, vol. 18, no. 1, pp. 26–29, Jun. 2019. DOI: 10.1109/LCA.2019.2893189

[265] U. Minhas, M. Russell, S. Kaloutsakis, P. Barber, R. Woods, G. Georgakoudis, C. Gillan, and D. S. Nikolopoulos, "NanoStreams: A Microserver Architecture for Real-Time Analytics on Fast Data Streams," ***IEEE Transactions on MultiScale Computing Systems***, vol. 4, no. 3, pp. 396–409, 2018.

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[263] C. Gillan, A. Novakovic, A. H. Marshall, M. Shyamsundar, and D. S. Nikolopoulos, "Expediting Assessments of Database Performance for Streams of Respiratory Parameters," ***Computers in Biology and Medicine***, vol. 100, no. 1, pp. 186–195, Sep. 2018.

[262] C. Reano, F. Silla, D. S. Nikolopoulos, and B. Varghese, "Intra-Node Memory Safe GPU Co-Scheduling," ***IEEE Transactions on Parallel and Distributed Systems***, vol. 29, no. 5, pp. 1089–1102, 2018.

[261] P. Thoman, K. Dichev, T. Heller, R. Iakymchuk, X. Aguilar, K. Hasanov, P. Gschwandtner, P. Lemarinier, S. Makridis, H. Jordan, T. Fahringer, K. Katrinis, E. Laure, and D. S. Nikolopoulos, "A Taxonomy of Task-Based Parallel Programming Technologies for High-Performance Computing," ***The Journal of Supercomputing***, vol. 74, no. 4, pp. 1422–1434, Apr. 2018.

[260] C. Chalios, G. Georgakoudis, K. Tovletoglou, G. Karakonstantis, H. Vandierendonck, and D. S. Nikolopoulos, "DARE: Data Access Aware Refresh via Spatio-Temporal Application Resilience on Commodity Servers," ***International Journal of High Performance Computing Applications***, vol. 32, no. 1, pp. 74–88, 2018. DOI: 10.1177/1094342017718612 [Online]. Available: http://journals.sagepub.com/doi/10.1177/1094342017718612

[259] G. Karakonstantis, D. S. Nikolopoulos, D. Gizopoulos, P. Trancoso, Y. Sazeides, C. D. Antonopoulos, S. Venugopal, and S. Das, "Error-Resilient Server Ecosystems for Edge and Cloud Datacenters," ***IEEE Computer***, vol. 50, no. 12, pp. 78–82, Dec. 2017.

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[257] G. Georgakoudis, H. Vandierendonck, P. Thoman, B. R. de Supinski, T. Fahringer, and D. S. Nikolopoulos, "SCALO: Scalability-Aware Parallelism Orchestration for Multi-Threaded Workloads," ***ACM Transactions on Architecture and Code Optimization***, vol. 14, no. 4, 54:1–54:25, 2017.

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[255] C.-H. Hong, I. T. A. Spence, and D. S. Nikolopoulos, "FairGV: Fair and Fast GPU Virtualization," ***IEEE Transactions on Parallel and Distributed Systems***, vol. 28, no. 12, pp. 3472–3485, Jun. 2017. DOI: 10.1109/TPDS.2017.2717908

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[246] G. Georgakoudis, C. J. Gillan, A. Sayed, I. Spence, R. Faloon, and D. S. Nikolopoulos, "Iso-Quality of Service: Fairly Ranking Servers for Real-Time Data Analytics," ***Parallel Processing Letters***, vol. 25, no. 3, 2015.

[245] A. Khasymski and D. S. Nikolopoulos, "Scalable Black-Box Prediction Models for Multi-Dimensional Adaptation on NUMA Multi-Cores," ***International Journal of Parallel, Emergent, and Distributed Systems***, vol. 30, no. 3, pp. 193–210, Apr. 2015.

[244] D. S. Nikolopoulos, H. Vandierendonck, N. Bellas, C. D. Antonopoulos, S. Lalis, G. Karakonstantis, A. Burg, and U. Naumann, "Energy Efficiency through Significance-Based Computing," ***IEEE Computer***, vol. 47, pp. 82–85, 7 Jul. 2014.

[243] I. Manousakis, F. S. Zakkak, P. Pratikakis, and D. S. Nikolopoulos, "TProf: An Energy Profiler for Task-Parallel Programs," ***Sustainable Computing: Informatics and Systems***, vol. 5, pp. 1–13, Mar. 2015.

[242] P. Gschwandtner, C. Chalios, D. S. Nikolopoulos, H. Vandierendonck, and T. Fahringer, "On the Potential of Significance-Driven Execution for Energy-Aware HPC," ***Computer Science – Research and Development***, vol. 30, no. 2, pp. 197–206, Feb. 2015.

[241] C. Symeonidou, P. Pratikakis, D. S. Nikolopoulos, and A. Bilas, "Distributed Region-Based Memory Allocation and Synchronization," ***International Journal of High Performance Computing Applications***, vol. 28, no. 4, pp. 406–414, Nov. 2014.

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[238] H. Vandierendonck, G. Tzenakis, and D. S. Nikolopoulos, “Analysis of Dependence Tracking Algorithms for Task Dataflow Execution,” ***ACM Transactions on Architecture and Code Optimization***, vol. 10, no. 4, pp. 1–24, Dec. 2013, Article No. 61.

[237] D. Li, B. R. D. Supinski, M. Schulz, D. S. Nikolopoulos, and K. W. Cameron, “Strategies for Energy Efficient Resource Management of Hybrid Programming Models,” ***IEEE Transactions on Parallel and Distributed Systems***, vol. 24, no. 1, pp. 144–157, Jan. 2013.

[236] I. Manousakis and D. S. Nikolopoulos, “EPC: A Power Instrumentation Controller for Embedded Applications,” ***ACM SIGBED Review***, vol. 9, no. 2, pp. 28–32, Jun. 2012.

[235] C.-Y. Su, D. Li, D. S. Nikolopoulos, M. Grove, K. W. Cameron, and B. R. D. Supinski, “Critical Path-Based Thread Placement for NUMA Systems,” ***ACM SIGMETRICS Performance Evaluation Review***, vol. 40, no. 2, pp. 106–112, Sep. 2012.

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[227] M. M. Rafique, B. Rose, A. R. Butt, and D. S. Nikolopoulos, "Supporting MapReduce on Asymmetric Multi-core Clusters," ***ACM SIGOPS Operating Systems Review***, vol. 43, no. 2, pp. 25–34, 2009.

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[219] D. S. Nikolopoulos, E. Artiaga, E. Ayguadé, and J. Labarta, "Scaling Non-Regular Shared-Memory Codes by Reusing Custom Loop Schedules," ***Scientific Programming***, vol. 11, no. 2, pp. 143–158, Apr. 2003.

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Papers in Archival Conference Proceedings

[213] J. Fan, Y. Zhang, X. Li, and D. S. Nikolopoulos, "Dllm-serve: Bridging the memory gap in diffusion language model serving," in *Proceedings of the 40th ACM International Conference on Supercomputing (**ICS**)*, Belfast, Northern Ireland, UK, 2026.

[212] A. Nafisi, V. Cvetkovic, C. Shin, K. Reece, K. Reddy, B. Torres, S. Heidari, M. Ellis, and D. S. Nikolopoulos, "Marco: Multi-agent code optimization for high-performance computing," in *ACM Capital Region Celebration of Women in Computing Conference **(CAPWIC)***, Mar. 2026.

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Theses

[2] D. S. Nikolopoulos, "System Software Support for Reducing Memory Latency on CC-NUMA Architectures," PhD Dissertation, Department of Computer Engineering and Informatics, University of Patras, Dec. 2000.

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Publication Metrics

Citation metrics (as of April 2026)

Google Scholar	7,757	42	146
Top 2% of Computer Scientists Worldwide according to Scopus Standardized Citation Indicators			

Knowledge and Technology Transfer & Commercialization

Consulting Activities

[3] Consulting expert, Labaton, undisclosed dispute. January 2026–present.

[2] Consulting expert, Taylor Wessing, undisclosed patent dispute. January 2026–present.

[1] Technology Consultant on Container/Enterprise-scale Datacenters, servescale.ai. November 2025–present.

Patents

Research on memory management for stream processing and in-memory database systems, supported by NSF grants 0715051,0720763, 0904844, led to foundational results later extended into U.S. Patents (1) 9,112,666 by former student Scott Schneider, and (2) 10,365,997, (3) 10,387,127, (4) 10,474,557, (5) 10,698,732 by former student Ahmad Hassan. In accordance with NSF and other federal funding agency policies on open dissemination, I declined to participate in inventorship but contributed to the original publication. Patent details below:

[1] "Elastic Auto-Parallelization for Stream Processing Applications based on Measured Throughput and Congestion," US Patent 9,112,666, Granted Aug. 18, 2015.

[2] "Optimizing DRAM memory based on read-to-write ratio of memory access latency," US Patent 10,365,997, Granted Jul. 30, 2019.

[3] "Detecting sequential access data and random access data for placement on hybrid main memory for in-memory databases," US Patent 10,387,127, Granted Aug. 20, 2019.

[4] "Source code profiling for line-level latency and energy consumption estimation," US Patent 10,474,557, Granted Nov. 12, 2019.

[5] "Page Ranking in Operating System Virtual Pages in Hybrid Memory Systems," US Patent 10,698,732, Granted Jun. 30, 2020.

Technology Transfer Activities

[1] Train-less deep neural network architecture search **IBM Watson Studio** (2019)

[2] Query planning and data placement for in-memory transactional-analytical databases **SAP** (2019)

[3] High-level program parallelization software for FPGAs **Crevinn** (2019)

[4] Query planning and data placement for in-memory transactional-analytical databases **SAP** (2019)

[5] Automatic scaling of streaming operators, **IBM** (2010)

[6] Adoption of methods for memory management, dynamic software scaling by **Samsung, Red-Hat, Microsoft, Oracle, IBM, Alpine Electronics, Thales, Reservoir Labs** (2008–today)

Research Grants

Funding summary (as of April 2026)
Total grant value across 53 awards: **$139.9M**
PI share (personal portion as PI/Co-PI): **$34.3M**
Currency conversion at year-of-award annual average rates. Major sponsors: NSF, DOE, EU Horizon 2020 & FP7, EPSRC, Royal Society, Marie Curie, IBM, Intel, SAP, Sony, Cisco.

[53] **[05/2026–05/2027] SCALE-R: Semantic Co-rendering & Adaptive Link-aware Encoding for Real-time Graphics**. Sony. PI. $150,000 total (your share: $75,000).

[52] **[10/2025–10/2026] Cloud–Edge–Neuromorphic Computing Integration for Multimodal Sensing and AI-Driven Modeling of Complex Biological Systems**. Virginia Tech Institute for Critical Technology & Applied Science / Institute for Advanced Computing. Co-PI. $50,000 total (your share: $12,500).

[51] **[08/2025–08/2026] Advancing AI-Driven Autonomous Experimentation for Turbulent Flows**. Virginia Tech College of Engineering Major Grant Initiative Program. PI. $50,000.

[50] **[05/2024–11/2024] GPU-Accelerated High-Performance Computing to Supercharge AI Models for Pandemic Prevention**. National Science Foundation (National AI Resource). Co-PI. HPC resource allocation (no $ amount).

[49] **[10/2023–09/2026] SWEET: Hardware and Software for Sustainable Wearable Edge Intelligence**. National Science Foundation (#2315851). PI. $600,000.

[48] **[08/2023–07/2024] WASML: Enabling Scalable Serverless Computer Vision**. Sony Faculty Innovation Award (#VTF-AG3ZURVF). PI. $100,000.

[47] **[02/2022–01/2023] VirtEdgeAI: Seamless Virtualization of AI Accelerators in Edge Computing Substrates**. CISCO (#VTF-446663). PI. $100,000.

[46] **[06/2021–05/2024] Collaborative Research: CNS Core: Medium: HardLambda: A new FaaS Abstraction for Cross-Stack Resource Management in Disaggregated Datacenters**. National Science Foundation (#2106634). Co-PI. $600,000 (your share: $210,000).

[45] **[12/2017–12/2018] Belfast Region City Deal, Queen's Global Innovation Institute**. UK Treasury, 77%, Belfast City Council 23%. PI. £56,000,000 (your share: £15,000,000).

[44] **[01/2018–06/2021] Biohaviour: Building the Blind Watchmaker**. EPSRC (#EP/R003564/1). Co-PI. £778,226 (your share: £194,556).

[43] **[08/2017–07/2018] Variability in Computing Systems**. Royal Academy of Engineering, Distinguished Visiting Fellowships. Host PI. £4,100.

[42] **[01/2017–12/2020] Scalable, Virtualized Data Center Acceleration**. Intel. Co-PI. £3,945 (your share: £795).

[41] **[01/2017–12/2020] OPRECOMP: Open Transprecision Computing**. EU, Horizon 2020 (#H2020-732631). PI. €5,999,510 (£5,091,933) (your share: €705,625 (£599,781)).

[40] **[02/2016–01/2019] UNISERVER: A Universal Micro-server Ecosystem by Exceeding the Energy and Performance Scaling Boundaries**. EU, Horizon 2020 (#H2020-688540). Co-PI. €4,815,810 (£4,333,717) (your share: €322,648 (£222,516)).

[39] **[02/2016–01/2019] VINEYARD: Versatile Integrated Accelerator-based Heterogeneous Datacenters**. EU, Horizon 2020 (#H2020-687628). PI. €6,283,895 (£4,467,972) (your share: €663,625 (£471,850)).

[38] **[10/2015–09/2020] Principles and Practice of Near Data Computing**. Royal Society Wolfson Research Merit Award (#WM150009). PI. £50,000.

[37] **[09/2015–09/2018] Meeting the Future Challenges of Heterogeneous and Extreme Scale Parallel Computing**. SFI-DEL, Investigator Awards (#14/IA/2474). PI. £521,947.

[36] **[10/2015–10/2018] ECOSCALE: Energy Efficient Heterogeneous Computing at Exascale**. EU, Horizon 2020 (#H2020-671632). PI. €4,237,398 (£2,922,346) (your share: €696,750 (£480,518)).

[35] **[10/2015–10/2018] ALLScale: An Exascale Programming, Multi-objective Optimization and Resilience Management Environment Based on Nested Recursive Parallelism**. EU, Horizon 2020 (#H2020-671603). PI. €3,366,196 (£2,463,217) (your share: €438,578 (£320,930)).

[34] **[08/2015–09/2017] Crevinn Teoranta OpenCL Knowledge Transfer**. Knowledge Transfer Partnership / Intertrade Ireland Fusion Project. PI. £39,000.

[33] **[03/2015–03/2018] SERT: Scale-Free, Energy-Efficient and Resilient CSE Software for Mega-Core Systems**. EPSRC (Software for the Future II) (#EP/M01147X/1). PI. £963,929 (your share: £694,909).

[32] **[01/2015–01/2018] RAPID: Heterogeneous Secure Multi-level Remote Acceleration Service for Low-Power Integrated Systems and Devices**. EU, Horizon 2020 (#H2020-644312). PI. €2,203,800 (£1,695,231) (your share: €326,925 (£251,481)).

[31] **[01/2015–01/2018] DIVIDEND: Distributed Heterogeneous Vertically Integrated Energy Efficient Data Centers**. EPSRC, CHIST-ERA (#EP/M015742/1). PI. €1,346,885 (£1,077,508) (your share: €279,646 (£223,717)).

[30] **[10/2014–10/2017] HPDCJ: Heterogeneous Parallel and Distributed Computing with Java**. EPSRC, CHIST-ERA (#EP/M015750/1). PI. €1,721,010 (£1,376,808) (your share: €178,159 (£142,527)).

[29] **[03/2014–03/2017] ASAP: An Adaptive, Highly Scalable Analytics Platform**. European Commission, FP7-ICT (#FP7-619706). Co-PI. €2,245,128 (£1,909,122) (your share: €407,720 (£346,701)).

[28] **[01/2014–02/2014] US-Ireland R&D Partnership Planning Grant: Cloud-based Electronic Integration of Patient Records (CLEAR)**. (#PG20). PI. £1,356.

[27] **[10/2013–10/2016] ENPOWER: Energy-Proportional Computing with Heterogeneous and Reconfigurable Processors**. EPSRC (#EP/L004232/1). PI. £741,043 (your share: £348,325).

[26] **[09/2013–09/2016] ALEA: Abstraction-Level Energy Accounting and Optimization in Many-Core Programming Languages**. EPSRC, System Approaches to Distributed and Embedded Architectures (#EP/L000555/1). PI. £669,561 (your share: £394,025).

[25] **[09/2013–09/2016] NanoStreams: A Hardware and Software Stack for Real-Time Analytics on Fast Data Streams**. European Commission, FP7-ICT, Objective 3 (#FP7-610509). PI. €3,300,000 (your share: €723,565).

[24] **[10/2013–10/2016] CACTOS: Context-Aware Cloud Topology Optimization and Simulation**. European Commission, FP7-ICT, Objective 1 (#FP7-610811). PI. €3,215,751 (your share: €583,330).

[23] **[08/2013–08/2019] SAP: PhD Project on High Availability for Petascale Systems**. SAP AG (#UK-2013-009). PI. £12,436.

[22] **[06/2013–06/2016] SCORPIO: Significance-Based Computing for Reliability and Power Optimization**. European Commission, FP7-FET-Open (#FP7-323872). PI. €1,890,775 (your share: €273,400).

[21] **[04/2013–04/2015] NovoSoft: Software Management of Non-Volatile Memory Hierarchies**. European Commission, Marie Curie Intra-European Fellowship (#FP7-327744). Scientist in Charge. €309,235.

[20] **[03/2013–03/2016] Characterizing and Optimizing In-Memory Database Systems for Emerging Memory Technologies**. SAP UK Limited (#R502). Co-PI. £34,298 (your share: £17,149).

[19] **[09/2012–09/2015] GEMSCLAIM: Greener Mobile Systems by Cross Layer Integrated Energy Management**. EPSRC, CHIST-ERA (#EP/K017594/1). PI. €1,776,688 (your share: €436,884).

[18] **[11/2012–06/2013] Exascale Mesh Generation Runtime Systems**. Royal Academy of Engineering, Distinguished Visiting Fellowships. Host PI. £4,100.

[17] **[01/2012–01/2016] HOLISTIC: Hardware and Software Techniques for Multicore Processor Architectures Reliability Enhancement**. Greek Ministry of Education, Lifelong Learning and Religious Affairs, Thales Programme (#1103). PI. €600,000 (your share: €98,000).

[16] **[03/2010–03/2012] SCC–MR: Scalable and Energy-Efficient Runtime Support for the MapReduce Programming Model on the Intel SCC**. Intel Corporation. PI. Equipment donation.

[15] **[06/2010–09/2012] TEXT: Towards Exascale Applications**. European Commission, FP7-INFRASTRUCTURES Programme (#ICT-261580). PI. €2,470,000 (your share: €299,364).

[14] **[06/2010–06/2011] ReMap: Rearchitecting MapReduce for Multicore Systems with Explicit Communication**. High Performance and Embedded Architectures and Compilers Network of Excellence, Cluster Collaboration Grant (#ICT-217068). PI. €3,000 share.

[13] **[03/2010–03/2013] ENCORE: Enabling Technologies for a Programmable Many-core**. European Commission, FP7-ICT, Objective 3 (#ICT-248647). Co-PI. €2,533,000 (your share: €533,000).

[12] **[01/2009–01/2013] I-Cores: Hypervisor-based Synthesis of Custom Execution Environments for Multi-core Systems**. European Commission, FP7 Programme, Marie Curie International Reintegration Grants (#IRG-224759). PI. €100,000.

[11] **[01/2008–02/2008] HiPEAC Fellowship: Runtime Systems for Parallel Programming**. European Commission, FP7 Programme, European Network of Excellence in High Performance and Embedded Architectures (#ICT-217068). PI. €8,600 share.

[10] **[07/2007–07/2010] VT-ASOS: Virtualization Technologies for Application-Specific Operating Systems on Many-Core HPC Systems**. NSF Computer Systems Research Program (#CNS-0720673). PI. $300,000 (your share: $150,000).

[9] **[07/2007–07/2010] Thermal Conductors: Runtime Software Support for Proactive Heat Management in Advanced Execution Systems**. NSF Computer Systems Research Program (#CNS-0720750). Co-PI. $350,000 (your share: $175,000).

[8] **[05/2007–05/2008] Models and Adaptive Runtime Systems for Accessible Parallel Programming on IBM Multi-Core Systems**. IBM Faculty Award Program (#VTF-874197). PI. $15,000.

[7] **[07/2007–07/2008] MISER: A High-Performance, Power-Aware Cluster**. NSF Computing Research Infrastructure Program (#CNS-0709025). Co-PI. $500,000 (your share: $166,667).

[6] **[08/2006–08/2007] Faculty Startup Grant**. Virginia Tech. PI. $100,000.

[5] **[08/2005–08/2008] MELISSES: Liquid Services for Scalable Multithreaded and Multicore Execution on Emerging Supercomputers**. DOE Early Career Principal Investigator Award Program (#DE-FG02-06ER25751, DE-FG02-05ER25689). PI. $299,907.

[4] **[05/2005–05/2008] Acquisition of STEMS: A Laboratory for End-to-End Development of Software and Tools for Emerging Multigrain Supercomputers**. NSF Major Research Instrumentation Program (#CNS-0521381). PI. $228,134 (your share: $76,045).

[3] **[05/2005–08/2005] A Unified Framework for Multilevel Parallelization on Deep Computing Systems**. NSF Research Experiences for Undergraduates Program (#CCF-0531887). PI. $6,000.

[2] **[01/2004–01/2009] A Unified Framework for Multilevel Parallelization on Deep Computing Systems**. NSF CAREER Award Program (#CCF-0346867, CCF-0715051). PI. $419,835 (your share: $225,000).

[1] **[08/2002–08/2004] Faculty Startup Grant**. College of William and Mary. PI. $100,000.

Education

Courses Taught

[43] Spring'26 CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: , Class Size 200)

[42] Spring'25 CS5914: Special Topics in Computer Science: LLMs for High-Performance Code Generation, Virginia Tech (Teaching Effectiveness: 5.70/6.00, Class Size 20)

[41] Fall'24 CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: 4.90/6.00, Class Size 195)

[40] Spring'24 CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: 5.10/6.00, Class Size 150)

[39] Fall'23 CS5510/ECE5510: Multiprocessor Programming, Virginia Tech (Teaching Effectiveness: N/A, Class Size 45)

[38] Spring'23 CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: 4.87/6.00, Class Size 192)

[37] Fall'22 CS5914: Warehouse Scale Computing, Virginia Tech (Teaching Effectiveness: 5.79/6.00, Class Size 17)

[36] Spring'22 CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: 5.16/6.00, Class Size 216)

[35] Fall'21 CS5510/ECE5510: Multiprocessor Programming, Virginia Tech (Teaching Effectiveness: 5.78/6.00, Class Size 35)

[34] Spring'21 CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: 5.15/6.00, Class Size 148)

[33] Fall'20 CS4284 Systems & Networking Capstone, Virginia Tech (Teaching Effectiveness: 5.85/6.00, Class Size: 23)

[32] Spring'20 CS2506: Computer Organization II, Virginia Tech (Teaching Effectiveness: 5.16/6.00, Class Size: 82)

[31] Spring'15 ECS 1002: Design Projects, Queen's University of Belfast

[30] Fall'15 ECS 2001: Second Stage Design Projects, Queen's University of Belfast

[29] Spring'14 ECS 1002: Design Projects, Queen's University of Belfast

[28] Fall'13 ECS 2001: Second Stage Design Projects, Queen's University of Belfast

[27] Spring'13 ECS 1002: Design Projects, Queen's University of Belfast

[26] Fall'12 ECS 2001: Second Stage Design Projects, Queen's University of Belfast

[25] Fall'11 CS425: Computer Systems Architecture, University of Crete

[24] Fall'11 CS100: Introduction to Computer Science, University of Crete

[23] Spring'11 CS225: Computer Organization, University of Crete

[22] Spring'11 CS529: Multi-core Systems Programming, University of Crete

[21] Fall'10 CS425: Computer Systems Architecture, University of Crete

[20] Spring'10 CS225: Computer Organization, University of Crete

[19] Spring'10 CS529: Multi-core Systems Programming, University of Crete

[18] Fall'09 CS425: Computer Systems Architecture, University of Crete

[17] Spring'09 CS529: Multi-core Systems Programming, University of Crete

[16] Spring'09 CS225: Computer Organization, University of Crete

[15] Fall'08 CS425: Computer Systems Architecture, University of Crete

[14] Spring'08 CS425: Computer Systems Architecture, University of Crete

[13] Fall'07 CS5234: Advanced Parallel Computation, Virginia Tech

[12] Fall'07 CS2504: Introduction to Computer Organization, Virginia Tech

[11] Spring'07 CS2504: Introduction to Computer Organization, Virginia Tech

[10] Fall'06 CS4234: Parallel Computation, Virginia Tech

[9] Spring'06 CSCI644: Advanced Computer Architecture, College of William and Mary

[8] Fall'05 CSCI444/544: Principles of Operating Systems, College of William and Mary

[7] Spring'05 CSCI644: Advanced Computer Architecture, College of William and Mary

[6] Fall'04 CSCI444/544: Principles of Operating Systems, College of William and Mary

[5] Spring'04 CSCI644: Advanced Computer Architecture, College of William and Mary

[4] Fall'03 CSCI444/544: Principles of Operating Systems, College of William and Mary

[3] Spring'03 CSCI644: Advanced Computer Architecture, College of William and Mary

[2] Fall'02 CSCI444/544: Principles of Operating Systems, College of William and Mary

[1] Spring'02 ECE291: Computer Engineering II, University of Illinois at Urbana-Champaign

Seminars Taught

Spring'12 Implementation of Multi-core Programming Models, Universitat Politecnica de Catalunya
Spring'10 Multi-core Systems Programming and Optimization, Universitat Politecnica de Catalunya
Spring'08 Multi-core Systems Programming and Optimization, Universitat Politecnica de Catalunya
Spring'07 Multi-core Systems Programming and Optimization, Universitat Politecnica de Catalunya
Spring'04 Multithreaded Architectures and Software, College of William and Mary

Course and Curriculum Development

CS2506: Intro to Computer Organization II, Redeveloped to introduce new open-source processor and tools, Virginia Tech, 2022
ECS2001: Software and Electronic Systems Engineering Design Projects (2nd Stage), Developed from scratch, Queen's University of Belfast, 2016
ECS1002: Software and Electronic Systems Engineering Design Projects (1st Stage), Developed from scratch, Queen's University of Belfast, 2016
CS529: Multicore Processor Programming, Developed from scratch, University of Crete, 2009
CS425: Computer Systems Architecture, Major revision, University of Crete, 2009
CS225: Computer Organization, Major revision (multi-core, cache coherence), University of Crete, 2009
CS5234: Advanced Parallel Computation, Developed from scratch, Virginia Tech, 2006

Teaching Grants

[4] **Advanced Topics in Implementing Multicore Programming Models**. Sponsor: Universitat Politecnica de Catalunya. Amount: €2,400 Role: PI. Dates of activity: 05/2012–06/2012.

[3] **Multi-core Systems Programming**. Sponsor: Universitat Politecnica de Catalunya. Amount: €2,400 Role: PI. Dates of activity: 05/2010–06/2010.

[2] **Multi-core Systems Programming**. Sponsor: Universitat Politecnica de Catalunya. Amount: €3,600. Role: PI. Dates of activity: 05/2008–06/2008.

[1] **Multi-core Systems Programming and Optimization**. Universitat Politecnica de Catalunya. Funding amount: €3,600. Role: PI. Dates of activity: 05/2007–06/2007.

Supervision

Ph.D. Thesis Students

Primary Advisor

[19] **Mona Moghadampanah** – Computer Science, Virginia Tech. In progress. Thesis area: *Scheduling of Workloads in the Continuum*.

[18] **Adib Rezaei** – Computer Science, Virginia Tech. In progress. Thesis area: *Accelerator Virtualization and Fault Tolerance*.

[17] **Sina Heidari** – Computer Science, Virginia Tech. In progress. Thesis area: *Large-Scale Processing of Graph Neural Networks*.

[16] **Jiakun Fan** – Computer Science, Virginia Tech. In progress. Thesis area: *Edge AI*.

[15] **Yunqi Shen** – Computer Science, Virginia Tech. In progress. Thesis area: *Tiered Memory Systems*.

[14] **Farhana Amin** – Computer Science, Virginia Tech. In progress. Thesis area: *Programming Models for and by LLMs*.

[13] **Xiangchen Li** – Electrical and Computer Engineering, Virginia Tech. In progress. Thesis area: *Edge Computing and Networking for AI*.

[12] **Dr. Emadeldin Abdrabou** – Electrical and Computer Engineering, Virginia Tech. December 2025. Thesis title: *On the Optimization of Edge Server Scaling and Placement for Open Radio Access Network (O-RAN) Slicing*. (co-advised with Scott Midkiff)

[11] **Dr. Esha Barlaskar** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast (co-supervised with Ivor Spence, Peter Kilpatrick). November 2020. Thesis title: *User-Centric Cloud Application Management*.

[10] **Dr. Nan Wang** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast. September 2019. Thesis title: *Resource Management for Edge Computing Systems*.

[9] **Dr. Roxana Istrate** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast. July 2019. Thesis title: *Efficient Neural Network Architecture Search*.

[8] **Dr. Sakil Barbhuiya** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast (co-supervised with Peter Kilpatrick). July 2018. Thesis title: *Anomaly Detection in Cloud and Mobile Devices*.

[7] **Dr. Charalambos Chalios** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast (co-supervised with Hans Vandierendonck). July 2017. Thesis title: *Software-Defined Significance-Based Computing*.

[6] **Dr. Giorgis Georgakoudis** – Computer and Telecommunications Engineering, University of Thessaly (co-supervised with Spyros Lalis). May 2016. Thesis title: *Scheduling and Performance Characterization on Heterogeneous Computing Systems*.

[5] **Dr. Jae-seung Yeom** – Computer Science, Virginia Tech (co-supervised with Madhav Marathe). May 2014. Thesis title: *Optimizing Data Accesses for Scaling Data-intensive Scientific Applications*.

[4] **Dr. Spyros Lyberis** – Computer Science, University of Crete. July 2013. Thesis title: *Myrmics: A Scalable Runtime System for Global Address Spaces*.

[3] **Dr. Scott Schneider** – Computer Science, Virginia Tech. December 2010. Thesis title: *Shared Memory Abstractions for Heterogeneous Multicore Processors*.

[2] **Dr. Filip Blagojevic** – Computer Science, Virginia Tech. May 2008. Thesis title: *Scheduling on Asymmetric Parallel Architectures*.

[1] **Dr. Matthew Curtis-Maury** – Computer Science, Virginia Tech. March 2008. Thesis title: *Improving the Efficiency of Parallel Applications on Multithreaded and Multicore Systems*. **Virginia Tech Outstanding Ph.D. Dissertation Award**.

Co-Advisor

[17] **Shunyu Yao** – Computer Science, Virginia Tech. In progress, co-advised with Ali R. Butt. *Serverless HPC*.

[16] **Yuze Li** – Computer Science, Virginia Tech. *Programming Language Support for Far Memory*.

[15] **Kazi Hassan Ibn Arif** – Computer Science, Virginia Tech. *Sparsity in Multimodal Learning*.

[14] **Dr. Abdullahi Abubakar** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast. December 2023. Thesis title: *Anomaly Detection in Longitudinal Data with Applications in Cloud Computing and Healthcare*.

[13] **Dr. Ioannis Tsiokanos** – Electronics, Electrical Engineering and Computer Science, Queen's University Belfast (co-supervised with Georgios Karakonstantis). December 2021. Thesis title: *Cross-layer Instruction-Aware Timing Error Mitigation & Evaluation for Energy Efficient Dependable Architectures*.

[12] **Dr. Konstantinos Tovletoglou** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast (co-supervised with Georgios Karakonstantis). June 2021. Thesis title: *Modeling and design of energy-efficient dependable memory sub-systems*.

[11] **Dr. Jiawen Sun** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast (co-supervised with Hans Vandierendonck). July 2017. Thesis title: *The GraphGrind framework: fast graph analytics on large shared-memory systems*.

[10] **Dr. Stuart McCool** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast (co-supervised with Peter Kilpatrick). July 2016. Thesis title: *Guidance Environments for Program Parallelization and Analysis*.

[9] **Dr. Ahmad Hassan** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast (co-supervised with Hans Vandierendonck). July 2016. Thesis title: *Software Management of Hybrid Main Memory Systems*.

[8] **Dr. Eoghan O'Neill** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast (co-supervised with Peter Kilpatrick). October 2015. Thesis title: *A Framework for Managing Shared Accelerators in Heterogeneous Environments*.

[7] **Dr. Aleksandr Khasymski** – Computer Science, Virginia Tech (co-supervised with Ali R. Butt). February 2015. Thesis title: *Accelerated Storage Systems*.

[6] **Dr. Chun-Yi Su** – Computer Science, Virginia Tech (co-supervised with Kirk W. Cameron). December 2014. Thesis title: *Energy-Aware Thread and Data Management in Heterogeneous Multi-Core and Multi-Memory Systems*.

[5] **Dr. Vassilis Papaefstathiou** – Computer Science, University of Crete (co-supervised with Manolis Katevenis). November 2013. Thesis title: *Architectural Support for Software-Guided Energy Reduction of Manycore Communication*.

[4] **Dr. Muhammad Mustafa Rafique** – Computer Science, Virginia Tech. September 2011 (co-supervised with Ali R. Butt). Thesis title: *An Adaptive Framework for Managing Heterogeneous Many-Core Clusters*.

[3] **Dr. Dong Li** – Computer Science, Virginia Tech. January 2011 (co-supervised with Kirk W. Cameron). *Scalable and Energy Efficient Execution Methods for Multicore Systems*.

[2] **Dr. John Christian Linford** – Computer Science, Virginia Tech (co-supervised with Adrian Sandu). May 2010. Thesis title: *Accelerating Atmospheric Modeling Through Emerging Multi-core Technologies*.

[1] **Dr. Richard Tran Mills** – Computer Science, College of William & Mary (co-supervised with Andreas Stathopoulos). November 2004. Thesis title: *Dynamic Adaptation to CPU and Memory Load in Scientific Applications*.

Postdoctoral Research Fellows

[17] **Dr. Dimitris Spatharakis** – Visiting Postdoctoral Fellow, School of Electrical and Computer Engineering, National Technical University of Athens. 04/25–05/25.

[16] **Dr. Sakil Barbhuiya** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast. Research themes. Cloud-based Distributed Manufacturing. 05/18–08/19. Research themes: Smart manufacturing as a service.

[15] **Dr. Jun-Kyu Lee** – EEECS, Queen's University of Belfast. 04/17–12/20. Research themes: Approximate computing.

[14] **Dr. Damon Fenacci** – EEECS, Queen's University of Belfast. 01/17–01/18. Research themes: Memory management.

[13] **Dr. Giorgis Georgakoudis** – EEECS, Queen's University of Belfast. 05/16–09/18. Research themes: System software and hardware/software interface.

[12] **Dr. Blesson Varghese** – EEECS, Queen's University of Belfast. 01/16–06/17. Research themes: Energy-efficient and resilient high-performance computing.

[11] **Dr. Kiril Dichev** – EEECS, Queen's University of Belfast. 10/15–08/19. Research themes: Exascale resilience.

[10] **Dr. Cheol-Ho Hong** – EEECS, Queen's University of Belfast. 06/15–12/17. Research themes: Accelerator virtualization.

[9] **Dr. Lev Mukhanov** – EEECS, Queen's University of Belfast. 04/14–12/20. Research themes: Abstraction-level energy accounting in many-core programming languages.

[8] **Dr. Zafeirios Papazachos** – EEECS, Queen's University of Belfast. 01/14 – 08/19. Research themes: Data center performance & reliability monitoring and optimization.

[7] **Dr. Hemant Mehta** – EEECS, Queen's University of Belfast. 10/15–04/17. Research themes: Virtual machine energy accounting.

[6] **Dr. Yun Wu** – EEECS, Queen's University of Belfast. 01/14–01/17. Research themes: Energy-proportional many-core computing systems.

[5] **Dr. Paul Harvey** – EEECS, Queen's University of Belfast. 01/16–09/16. Research themes: Exascale programming models.

[4] **Dr. Ahmed Sayed** – EEECS, Queen's University of Belfast. 09/14–02/15. Research themes: System software for real-time in-memory analytics.

[3] **Dr. Konstantina Mitropoulou** – EEECS, Queen's University of Belfast. 01/14–03/14. Research themes: System software for real-time in-memory analytics.

[2] **Dr. Hans Vandierendonck** – FORTH-ICS. 10/10–10/11. Research themes: Parallel programming, scheduling.

[1] **Dr. Christos D. Antonopoulos** – Computer Science, College of William & Mary. 06/04–06/06. Research themes: Energy-efficient parallel computation, runtime systems, and memory management.

Pre-Doctoral Research Assistants

[4] **Kai Chen** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast. In progress. Research area: *Multi-scale Power and Performance Modeling*.

[3] **George Tzenakis** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast. Research area: *Dynamic Parallelism and Elasticity* 10/12–02/17.

[2] **Chhaya Trehan** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast. Research area: *Optimization Algorithms for Energy-Efficiency* 01/15–05/16.

[1] **Mahwish Arif** – Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast (co-supervised with Hans Vandierendonck). Research area: *Performance Portability*.

Visiting Scholars

[2] **Prof. Oscar Garcia Lorenzo** – Department of Computer Architecture, University of Santiago de Compostela. April 2016. Thesis title: *Hardware counter-based performance analysis, modeling, and improvement through thread migration in NUMA systems*.

[1] **Dr. Satoshi Imamura** – System LSI Laboratory. Kyushu University. Research area: *Energy-Efficiency Optimization on Multicore NUMA Servers*.

M.Sc. Research Students

Primary Advisor

[29] **Katelyn Crumpacker** – Computer Science, Virginia Tech. *AI-Generated Energy-Efficient Software.*

[28] **Timothy Coyne** – Computer Science, Virginia Tech, May 2025. *Performance Portability of CUDA Across NVIDIA GPU Architectures*.

[27] **Max Fisher** – Computer Science, Virginia Tech, May 2025. Thesis title: *Analysis of Memory Access Patterns and Pre-Fetching Strategies for Large Language Model Inferencing* .

[26] **Moustafa Kahla** – Electrical and Computer Engineering, Virginia Tech. December 2023. Thesis title: *Automatic Source Code Transformation To Pass Compiler Optimization*.

[25] **Manthan Shah** – Electrical and Computer Engineering, Virginia Tech. June 2023. Thesis title: *Analysis of YOLOv7 Inference Performance through a Systems Perspective and the identification of optimization opportunities*.

[24] **Shreya Bhandare** – Computer Science, Virginia Tech. June 2023. Thesis title: *Designing RDMA-based Efficient Communication for GPU Remoting.*.

[23] **Melissa Cameron** – Computer Science, Virginia Tech. June 2023. Thesis title: *Parallel Islands: A Diversity Aware Tool For Parallel Computing Education*.

[22] **Ishaan Gulati** – Computer Science, Virginia Tech. June 2023. Thesis title: *A Scalable Leader-Based Consensus Algorithm*.

[21] **Matthew Jackson** – Computer Science, Virginia Tech. May 2023. Thesis title: *Computational Offloading for Real-Time Computer Vision in Unreliable Multi-Tenant Edge Systems*.

[20] **Daniel Moyer** – Computer Science, Virginia Tech. May 2021. Thesis title: *Punching Holes in the Cloud: Direct Communication between Serverless Functions using NAT Traversal*

[19] **Dimitris Chassapis** – Computer Science, University of Crete. Thesis title: *Static Analysis for Parallelism and Correctness in Task Dataflow Programming Models*.

[18] **Ioannis Manousakis** – Computer Science, University of Crete. July 2013. Thesis title: *TPROF: An Energy Profiler for Task-Parallel Programs*.

[17] **Evangelos Kafentarakis** – Computer Science, University of Crete. July 2013. Thesis title: *Lprof: A Tool for Profiling Locality Awareness in a Task-Based Programming Model*.

[16] **Christi Symeonidou** – Computer Science, University of Crete. July 2013. Thesis title: *Distributed Region-Based Allocation and Synchronization*.

[15] **Kallia Chronaki** – Computer Science, University of Crete. June 2013. Thesis title: *Exploiting Pipelined Parallelism with Task Dataflow Programming Models*.

[14] **Alexandros Labrineas** – Computer Science, University of Crete. June 2013. Thesis title: *BDDT-SCC: A Task-Parallel Runtime for the Single-Chip Cloud Computer*.

[13] **Anastasios Papagiannis** – Computer Science, University of Crete. March 2013. Thesis title: *MapReduce on Distributed-Memory Many-Core Architectures*.

[12] **Angelos Papatriantafyllou** – Computer Science, University of Crete. March 2012. Thesis title: *Optimized Block-Based Dependence Analysis for Task Parallelism*.

[11] **Constantinos Koukos** – Computer Science, University of Crete (co-supervised with Angelos Bilas). August 2010. Thesis title: *Locality Management in Task-Based Parallel Programming Models*.

[10] **Pranav Tendulkar** – ALaRi Institute Advanced Studies in Embedded Systems Design. June 2010. Thesis title: *Runtime OpenMP Support using Hardware Primitives on Explicitly Memory Managed Multi-Processors*.

[9] **Michail Zampetakis** – Computer Science, University of Crete. April 2010. *Runtime Support for Programming Explicit Communication Chip Multiprocessors*.

[8] **Maria Katsamani** – Computer Science, University of Crete (co-supervised with Manolis Katevenis). March 2010. Thesis title: *Software Implementation of MPI Primitives on Multicore FPGA*.

[7] **Benjamin Rose** – Computer Science, Virginia Tech. May 2009. Thesis title: *Intra- and Inter-Chip Communication Support for Asymmetric Multicore Processors with Explicitly Managed Memory Hierarchies*.

[6] **Beran Nova Bryant** – Computer Science, Virginia Tech. May 2008. *Temperature-Aware Scheduling of Parallel Applications on Shared-Memory Multiprocessors.*

[5] **Harshil Shah** – Computer Science, Virginia Tech. May 2008. *Application Parallelization on the Cell/BE.*

[4] **Jyotirmaya Tripathi** – Computer Science, Virginia Tech. May 2008. *Scheduling Parallel Applications on Paravirtualized Shared-Memory Multiprocessors*.

[3] **Ankur Shah** – Computer Science, Virginia Tech. April 2008. Thesis title: *Prediction Models for Multi-dimensional Power-Performance Optimization on Many Cores*.

[2] **Scott Schneider** – Computer Science, College of William & Mary. June 2005. Thesis title: *Factory: An Object-Oriented Parallel Programming Substrate for Deep Multiprocessors*.

[1] **Robert McGregor** – Computer Science, College of William & Mary. May 2005. *Scheduling with Bus Bandwidth Considerations on Shared-Memory Multiprocessors.*

Co-Advisor

[4] **Foivos Zakkak** – Computer Science, University of Crete (co-supervised with Angelos Bilas). March 2012. Thesis title: *SCOOP: Language Extensions and Compiler Optimizations for Task-based Programming Models*.

[3] **Ioannis Kesapides** – Computer Science, University of Crete (co-supervised with Angelos Bilas). March 2011. Thesis title: *Dynamic Dependence Analysis on Multi-core Processors*.

[2] **Michail Alvanos** – Computer Science, University of Crete (co-supervised with Angelos Bilas). June 2010. Thesis title: *Design and Evaluation of a Task-based Parallel H.264 Video Encoder for the Cell Processor*.

[1] **George Tzenakis** – Computer Science, University of Crete (co-supervised with Angelos Bilas). October 2009. Thesis title: *Tagged Procedure Calls (TPC): Efficient Runtime Support for Task-Based Parallelism on the Cell Processor.*

Undergraduate (BEng/BSc) Research Students

[43] **Kayden Moraes** – Computer Science, Virginia Tech. Fall 2025. *Agentic AI for HPC Code Optimization*.

[42] **Adarsh Chittimoori** – Computer Science, Virginia Tech. Fall 2025. *Agentic AI for HPC Code Optimization*.

[41] **Claire Shin** – Computer Science, Virginia Tech. Fall 2025. *Agentic AI for HPC Code Optimization*.

[40] **Ariana Nafisi** – Computer Science, Virginia Tech. Fall 2025. *Agentic AI for HPC Code Optimization*.

[39] **Krishnateja Reddy** – Computer Science, Virginia Tech. Spring 2025, Fall 2025. *Agentic AI for HPC Code Optimization*.

[38] **Kathleen Reece** – Computer Science, Virginia Tech. Spring 2025, Fall 2025. *Agentic AI for HPC Code Optimization*.

[37] **Bryan Torres** – Computer Science, Virginia Tech. Spring 2025, Fall 2025. *Agentic AI for HPC Code Optimization*.

[36] **Veljko Cvetkovic** – Computer Science, Virginia Tech. Spring 2025, Fall 2025. *Agentic AI for HPC Code Optimization*.

[35] **Yasir Hasan** – Computer Science, Virginia Tech. Fall 2024-Spring 2025. *Agentic AI for HPC Code Optimization*.

[34] **Denise Cooney** – Computer Science, Virginia Tech. Spring 2025. *Agentic AI for HPC Code Optimization*.

[33] **Asif Rahman** – Computer Science, Virginia Tech. Fall 2024-Spring 2025. *Agentic AI for HPC Code Optimization*.

[32] **Aneesh Tummeti** – Computer Science, Virginia Tech. Fall 2024-Spring 2025. *Agentic AI for HPC Code Optimization*.

[31] **Aidan Walters** – Computer Science, Virginia Tech. Fall 2024-Spring 2025. *Agentic AI for HPC Code Optimization*.

[30] **Jack Williamson** – Computer Science, Virginia Tech. Summer 2023, Fall 2023. *Supporting flexible SIMD ISAs in WebAssembly*.

[29] **Anthony Nguyen** – Computer Science, Virginia Tech. Summer 2023, Fall 2023. *WebAssembly Performance Profiling*.

[28] **Alex Lin** – Computer Science, Virginia Tech. Summer 2023, Fall 2023. *Supporting GPUs in WebAssembly*.

[27] **Gianfranco Vivanco** – Computer Science, Virginia Tech. Summer 2023, Fall 2023. *Performance Characterization of LLM Inference*

[26] **Tatiana Monteiro** – Computer Science, Virginia Tech. Summer 2023. *Optimization of Inference from Decision Trees*.

[25] **Jamie Whiting** – Computer Science, Virginia Tech. Summer 2023. *RISC-V Simulation and System Software*.

[24] **Riyos Pudasaini** – Computer Science, Virginia Tech. Summer 2023. *Scalable Memory Allocators for Rust*.

[23] **Angel Gonzalez** – Computer Science, Virginia Tech. Summer 2023. *Cache Simulators*.

[22] **Kieran Siek** – Computer Science, Virginia Tech. Spring 2022. *Code Generation for Matrix ISA Extensions*

[21] **Aditya Iyer** – Computer Science, Virginia Tech. Spring 2022. *FaaS Workload Characterization*

[20] **Parker Harnack** – Computer Science, Virginia Tech. Summer 2021. *Performance Analysis of Persistent Memory Allocators*.

[19] **Lalitha Kupa** – Computer Science, Virginia Tech. Summer 2021. *NUMA Persistent Memory Management in the Operating System*.

[18] **Ishaan Singh** – Computer Science, Virginia Tech. Summer 2020. *Container Caching*.

[17] **Nikolaos Parasyris** – Electrical and Computer Engineering, National Technical University of Athens. September 2015. *Fine-grain energy profiling of large software repositories*.

[16] **Stylianos Ninidakis** – Computer Science, University of Crete. June 2011. *Parallelizing Irregular applications with Task Dataflow*.

[15] **Nikolaos Papakonstantinou** – Computer Science, University of Crete. June 2011. *Distributed Dynamic Dependence Analysis for Task Dataflow Models.*

[14] **Nikolaos Papadopoulos** – Computer Science, University of Crete, February 2012. *Scheduler-Driven Dynamic Data Placement for NUMA Multi-cores.*

[13] **Ioannis Manousakis** – Computer Science, University of Crete, May 2011. *Component-level Power Instrumentation on Multiprocessors.*

[12] **Dimitrios Chassapis** – Computer Science, University of Crete, May 2011. *Static Dependence Analysis for Task Dataflow Models.*

[11] **Christi Symeonidou** – Computer Science, University of Crete, May 2011. *Multi-node Communication Layer on the SARC FPGA Prototype.*

[10] **Alexandros Labrineas** – Computer Science, University of Crete, May 2011. *Early Release Optimizations for Task Dataflow Programming Models.*

[9] **Kallia Chronaki** – Computer Science, University of Crete, May 2011. *Parallel Loop Scheduling on the SARC Multi-core Processor*.

[8] **Christos Margiolas** – Computer Science, University of Crete. June 2010. *Data Placement and NUMA-Aware Optimization of MapReduce.*

[7] **Foivos Zakkak** – Computer Science, University of Crete, June 2010 (co-supervised with Angelos Bilas). *Source-to-Source Compiler Optimizations for Task Parallelism.*

[6] **Spyros Tsatuhas** – Computer Science, University of Crete, June 2010. *POSIX Threads Library Implementation on the SARC FPGA Prototype.*

[5] **Evangelos Kafentarakis** – Computer Science, University of Crete, June 2009. *Software Shared Memory Layer for CPU-GPU Systems.*

[4] **Anastasios Papagiannis** – Computer Science, University of Crete, June 2009. *Performance Analysis of Virtual Machine Schedulers in Xen.*

[3] **Patric Fiaux** – Computer Science, Virginia Tech, May 2007. *Application Parallelization and Optimization on Cell/BE.*

[2] **James Dzierwa** – Computer Science, College of William & Mary, May 2006. *Hardware Monitors for Power-Performance Adaptation.*

[1] **Evan McCreedy** – Computer Science, College of William & Mary, May 2004. *Multi-level Parallelization of MPIBlast.*

Service

Professional Activities

Membership in Professional Societies

IEEE Computer Society (IEEE), Fellow (2024–present), Distinguished Visitor (2024–present), Distinguished Contributor (2022–present), Senior Member (2011–2022), Member (1995–2011)
Association for Computing Machinery (ACM), Distinguished Member (2018–present), Senior Member (2011–2018), Member (1995–2011)
Asia-Pacific Artificial Intelligence Association (AAIA), Fellow (2024–present)
International Artificial Intelligence Industry Alliance (AIIA), Fellow (2024–present)
British Computer Society (BCS), Fellow (2014–present)
The Institution of Engineering and Technology (IET), Fellow (2017–present)
United Kingdom Council of Professors and Heads of Computing (CPHC), Member (2012–2019)
ACM Special Interest Group on Computer Architecture (SIGARCH), Member (2000–present)
ACM Special Interest Group on Operating Systems (SIGOPS), Member (2000–present)
ACM Special Interest Group on High Performance Computing (SIGHPC), Member (2000–present)
Chartered Engineer (CEng) (2017–present)
Technical Chamber of Greece, Member (1996–present)

Conference Committee Activities

Steering Committee Chair:

[2] ACM International Conference on Supercomputing (ICS) – 2024 – present

[1] IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) – 2022 – 2023

Steering Committee Associate Chair:

[1] ACM International Conference on Supercomputing (ICS) – 2023 – 2024

Program Chair/Co-Chair:

[9] ACM International Conference on High Performance Parallel and Distributed Computing (HPDC) – 2026

[8] ACM International Conference on Supercomputing (ICS) – 2023

[7] ACM International Conference on Supercomputing (ICS) – 2022

[6] International Workshop on Deployment and Use of Accelerators (DUAC) – 2025, 2024, 2023, 2022, 2021

[5] IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID) – 2014

[4] International Workshop on Power-Aware Algorithms, Systems and Architectures (PASA) – 2013

[3] EuroMPI MPI Conference – 2011

[2] IEEE International Conference on Scalable Computing and Communications (ScalCom) – 2011

[1] Workshop on Parallel Programming for Accelerators (PPAC) – 2009, 2010, 2011

Program Vice Chair/Area Chair:

[7] IEEE International Parallel and Distributed Processing Symposium (IPDPS), Area Chair for Programming Models and Runtime Systems – 2026

[6] International European Conference on Parallel and Distributed Computing (EUROPAR), Global Area Chair, Scheduling and Load Management – 2022

[5] International Conference on Embedded Computer Systems: Architecture, Modeling and Simulation (SAMOS) – 2016

[4] IEEE/ACM International Conference on High-Performance Computing, Networking, Architecture and Storage (SC) – 2014

[3] European Conference on Parallel Processing (EuroPar) – 2012

[2] IEEE International Parallel and Distributed Processing Symposium (IPDPS) – 2011

[1] International Conference on Parallel Processing (ICPP) – 2007

General Chair:

[10] IEEE International Conference on Cluster Computing (CLUSTER) – 2018, 2010

[9] IEEE International Symposium on the Performance Analysis of Systems and Software (ISPASS) – 2018

[8] ISC Conference Workshop on Approximate and Transprecision Computing on Emerging Technologies – 2018

[7] HiPEAC Conference Workshop on Approximate Computing – 2018

[6] SC Conference Workshop on Energy Efficient Supercomputing – 2017, 2016, 2015, 2014, 2013

[5] ParCo Conference MiniSymposium on Edge Computing – 2017

[4] ParCo Conference MiniSymposium on Parallel Programming for Reliability and Energy Efficiency – 2016

[3] ParCo Conference MiniSymposium on Energy and Resilience in Parallel Programming – 2015

[2] HiPEAC Conference Workshop on Energy Efficient Heterogeneous Computing – 2015

[1] ICPP Conference Workshop on Power Aware Systems and Architectures – 2013

Program Committee:

[54] IEEE/ACM International Conference on High-Performance Computing, Networking, Storage and Analysis (SC) – 2026, 2018, 2016, 2015, 2013, 2012

[53] ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) – 2026, 2021

[52] Annual AAAI Conference on Artificial Intelligence (AAAI), Senior Program Committee Member – 2026

[51] ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP) – 2026, 2025, 2021, 2020, 2015, 2013

[50] International Conference on Parallel Processing (ICPP) – 2025, 2024, 2021, 2020, 2019, 2018, 2017, 2016, 2014, 2008, 2004, 2003

[49] IEEE/ACM International Symposium on Microarchitecture (Micro) – 2025, 2024 (External Review Committee)

[48] IEEE International Conference on Parallel Architectures and Compilation Techniques (PACT) – 2025, 2023, 2022, 2009

[47] International Symposium on Computer Architecture (ISCA) – 2025

[46] IEEE/ACM International Symposium on High Performance Distributed Computing (HPDC) – 2025, 2021, 2020, 2019

[45] ACM International Conference on Supercomputing (ICS) – 2025, 2021, 2020, 2017, 2014, 2012, 2011, 2009, 2007

[44] IEEE/SBC International Symposium on Computer Architecture and High-Performance Computing (SBAC-PAD) – 2025

[43] IEEE International Parallel and Distributed Processing Symposium (IPDPS) – 2024, 2023, 2021, 2020, 2019, 2018, 2017, 2014, 2013

[42] European Conference on Parallel Processing (EuroPar) – 2024, 2023, 2020, 2016, 2014, 2013

[41] IEEE Big Data – 2024, 2023, 2018, 2017, 2016, 2015, 2014, 2013

[40] IEEE International Conference on Edge Computing (EDGE) – 2022

[39] IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID) – 2022, 2021, 2020, 2019, 2018, 2016, 2015, 2013

[38] International Supercomputing Conference (ISC) – 2020, 2019, 2018

[37] International Green and Sustainable Computing Conference (IGSC) – 2020, 2019

[36] Workshop on Languages and Compilers for Parallel Computing (LCPC) – 2020

[35] IEEE International Conference on High-Performance Computing (HiPC) – 2019, 2016, 2015, 2014

[34] Parallel Computing Conference (ParCo) – 2019, 2017, 2015

[33] IEEE Graph Computing Conference (GC) – 2019

[32] ACM International Conference on Distributed Event-Based Systems (DEBS) – 2018

[31] International Conference on Embedded Computer Systems: Architecture, Modeling, and Simulation (SAMOS) – 2018, 2017, 2015, 2014

[30] ISC Conference Workshop on Communication Architectures for HPC, Big Data, Deep Learning, and Clouds at Extreme Scale (ExaComm) – 2018, 2017

[29] ACM Conference on Computing Frontiers (CF) – 2017, 2015, 2014, 2011

[28] IEEE Conference on Cluster Computing (CLUSTER) – 2016, 2015, 2013, 2012, 2011

[27] ISC Workshop on Energy-Aware High-Performance Computing (EnaHPC) – 2017, 2016, 2015, 2014

[26] IEEE International Conference on Big Data (BigData) – 2024, 2017, 2016, 2014

[25] IEEE International Conference on Ubiquitous Computing and Communications (IUCC) – 2016, 2012

[24] HiPEAC International Conference on High Performance and Embedded Architectures and Compilation – 2015, 2014, 2013, 2012

[23] ACM PPoPP Conference Workshop on General Purpose Processing using GPU (GPGPU) – 2015, 2014

[22] Feedback Computing Conference – 2015, 2010

[21] IEEE International Conference on Green Computing and Communications (GreenCom) – 2013, 2011, 2010

[20] IEEE International Conference on Parallel and Distributed Systems (ICPADS) – 2012, 2010, 2006, 2004

[19] IEEE High-Performance Computing and Communications Conference (HPCC) – 2012, 2009

[18] Symposium on Application Accelerators for High-Performance Computing (SAAHPC) – 2012, 2011, 2010, 2009

[17] IFIP Network and Parallel Computing Conference (NPC) – 2012, 2011, 2010

[16] International Conference on Architecture of Computer Systems (ARCS) – 2012, 2011, 2010

[15] IEEE Network and Storage Systems Conference (NAS) – 2011, 2010, 2009

[14] IEEE International Conference on e-Business Engineering (ICEBE) – 2011, 2010

[13] IEEE Green Computing Conference – 2010

[12] International Conference on Algorithms and Architectures for Parallel Processing (ICA3PP) – 2010

[11] IEEE International Conference on Scalable Computing and Communications (ScalCom) – 2010, 2009

[10] IEEE Cloud Computing Conference (CloudCom) – 2010, 2009

[9] IEEE International Conference on Computational Science and Engineering (CSE) – 2010, 2009

[8] International Conference on Future Computational Technologies and Applications (Future Computing) – 2010

[7] International Conference on Parallel and Distributed Computing, Applications and Technologies (PDCAT) – 2010

[6] IEEE Autonomic and Trusted Computing Conference (ATC) – 2008, 2007, 2006

[5] Balkan Conference on Informatics (BCI) – 2007

[4] ACM SIGMETRICS International Conference on Measurement and Modeling of Computer and Communication Systems – 2006

[3] IEEE International Conference on Pervasive Systems (ICPS) – 2005

[2] International Symposium on High Performance Computing (ISHPC) – 2003

[1] International Conference on Computational Science (ICCS) – 2001

Program Committee (Workshops):

[21] ICPP Conference Workshop on Heterogeneous and Unconventional Cluster Architectures and Applications (HUCAA) – 2016, 2015

[20] ParCo Conference MiniSymposium on Parallel Programming for Reliability and Energy-Efficiency (PP4REE) – 2016

[19] HiPEAC Conference Workshop on Approximate Computing (WAPCO) – 2016, 2015

[18] CGO Conference Workshop on Code Optimization for Multi- and Many-Cores (COSMIC) – 2015, 2014

[17] HiPEAC Conference Workshop on Programmability and Architectures for Heterogeneous Multicores (MULTIPROG) – 2015, 2014, 2013, 2012, 2011, 2010

[16] ICPP Conference Workshop on Parallel Programming Models and System Software for High-End Computing (P2S2) – 2014, 2013

[15] SC Conference Energy-Efficient Supercomputing Workshop (E2SC) – 2014, 2013

[14] ISC Conference Workshop on Virtualization for High-Performance Computing (VHPC) – 2014, 2013, 2012, 2011, 2010, 2009, 2008

[13] PLDI Conference Workshop on Memory Systems Performance and Correctness (MSPC) – 2014, 2011

[12] IPDPS Workshop on Accelerators and Hybrid Exascale Systems (ASHES) – 2014, 2013, 2012

[11] IBM Extreme Scale Parallel Architecture and Systems Workshop (ESPAS) – 2014

[10] Embedded Operating Systems Workshop (EWiLi) – 2013

[9] ISCA Conference Workshop on Future Architectural Support for Parallel Programming Workshop (FASPP) – 2012, 2011

[8] HiPEAC Workshop on Computer Architecture and Operating System CoDesign (CAOS) – 2012

[7] PACT Conference Workshop on Programming Models for Emerging Architectures (PMEA) – 2011, 2010, 2009

[6] IPDPS Conference Workshop on Software Engineering Innovation for HPC Clouds (SinHPC) – 2011

[5] ICS Conference on Workshop on Characterizing Applications for Heterogeneous Exascale Systems (CACHES) – 2011

[4] International Workshop on Cloud Computing Interoperability and Services (InterCloud) – 2010

[3] ISCA Conference Workshop on Next Generation Multi/Many-core Technologies (FMT) – 2010, 2008

[2] IPDPS Conference High-Performance Power-Aware Computing Workshop (HPPAC) – 2009, 2008

[1] IBM Cell Processor Programming Workshop (CELL) – 2009

Steering Committee Member:

[3] ACM International Conference on Supercomputing (ICS) – 2022–present

[2] IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) – 2018–present

[1] IEEE International Conference on Cluster Computing (CLUSTER) – 2009–2011, 2017–2019

Other Committee Service:

[9] IEEE/ACM International Symposium on Microarchitecture (MICRO) – External Review Committee, 2024

[8] IEEE/ACM International Conference on High-Performance Computing, Networking, Storage and Analysis (SC) – Reproducibility Challenge Committee 2021

[7] IEEE/ACM International Conference on High-Performance Computing, Networking, Storage and Analysis (SC) – HPC Impact Showcase Chair 2017. Tutorials Committee 2014, 2013

[6] ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP) – External Review Committee 2014, 2012

[5] International Supercomputing Conference (ISC) – Tutorials Committee 2014, 2013, 2012

[4] HiPEAC Conference – Workshop and Tutorials Chair 2011

[3] EuroMPI Conference – Finance Chair 2011

[2] ACM International Conference on Supercomputing (ICS) – Finance Chair 2009, Workshops and Tutorials Chair 2007

[1] SIAM Conference on Parallel Processing for Scientific Computing – Mini-Workshop Organizer 2006, 2004

Session Chair

[20] ACM International Conference on Supercomputing (ICS) – 2025

[19] ACM International Conference on Supercomputing (ICS) – 2023 (Keynote Sessions)

[18] ACM International Conference on Supercomputing (ICS) – 2022 (Keynote Sessions)

[17] Workshop on Deployment and use of Accelerators (DUAC) – 2023, 2022

[16] IEEE/ACM International Symposium on High-Performance Distributed Computing (HPDC) – 2021

[15] ACM International Conference on Supercomputing (ICS) – 2021

[14] ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) – 2021

[13] ACM International Conference on Supercomputing (ICS) – 2020

[12] IEEE/ACM International Symposium on High-Performance Distributed Computing (HPDC) – 2020

[11] ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) – 2020

[10] IPDPS Workshop on Variability in Computing Systems (VarSys) – 2016

[9] SC Workshop on Energy-Efficient Supercomputing (E2SC) – 2014

[8] IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID) – 2014

[7] International Conference on Embedded Computer Systems: Architecture, Modeling and Simulation (SAMOS) – 2013

[6] ICPP Conference Workshop on Power Aware Systems and Architectures – 2013

[5] IEEE/ACM International Conference on High-Performance Computing, Networking, Analysis and Storage (SC) – 2013, 2012

[4] ACM International Conference on Computing Frontiers (CF) – 2011

[3] International Conference on High Performance and Embedded Architectures and Compilation (HiPEAC) – 2011, 2008

[2] International Conference on Parallel Processing (ICPP) – 2010

[1] IEEE International Parallel and Distributed Processing Symposium (IPDPS) – 2009

Editorial Boards

[17] Computer Physics Communications. **HPC Specialist Editor**, 2017–present

[16] Journal of Computational Sciences. **Editorial Board Member**, 2014–present

[15] International Journal of High Performance Computing Applications (IJHPCA). **Associate Editor**, 2012–present

[14] Future Internet. **Editorial Board Member**, 2020–2026

[13] Frontiers in High-Performance Computing. **Editor**, 2023–2024

[12] IEEE Transactions on Parallel and Distributed Systems. **Associate Editor**, 2018–2021

[11] IEEE Transactions on Parallel and Distributed Systems. **Best Paper Committee**, 2021

[10] IEEE Transactions on Parallel and Distributed Systems. **Artifact Evaluation Committee**, 2020–2021

[9] International Journal of Parallel, Emergent and Distributed Systems (IJPEDS). **Associate Editor**, 2010–2019

[8] Sustainable Computing: Informatics and Systems (SUSCOM). **Editorial Board Member**, 2010–present

[7] International Journal of Information Technology, Communications and Convergence. **Editorial Board Member**, 2009

[6] Scientific Programming. **Editorial Board Member**, 2015–2016

[5] Concurrency and Computation: Practice and Experience (CCPE). **Editorial Review Board**, 2015

[4] Parallel Computing (PARCO). **Guest Editor**, 2015

[3] IET Computers and Digital Techniques. **Guest Editor**, 2014

[2] Sustainable Computing: Informatics and Systems (SUSCOM). **Guest Editor**, 2014

[1] Journal of Autonomic and Trusted Computing. **Editorial Board Member**, 2006–2007

Reviewer Work for Technical Journals, Conferences, and Publishers post 2020

[17] ACM Transactions on Autonomous and Adaptive Systems (2025 – 1 paper)

[16] IEEE Transactions on Services Computing (2025 – 1 paper)

[15] Future Generation Computer Systems (2025 – 1 paper)

[14] IEEE Transactions on Mobile Computing (2025 – 1 paper)

[13] Frontiers in High Performance Computing (2024 – 1 paper)

[12] Future Generation Computer Systems (2023 – 1 paper)

[11] Computer Physics Communications (2022 – 1 paper)

[10] Engineering with Computers (2022 – 1 paper)

[9] Software – Practice & Experience (2022 – 1 paper)

[8] Concurrency and Computation – Practice & Experience (2022 – 1 paper)

[7] International Journal of High-Performance Computing Applications (2022 – 1 paper)

[6] Future Generation Computer Systems (2021 – 1 paper)

[5] Journal of Parallel and Distributed Computing (2021 – 1 paper)

[4] IEEE Access (2021 – 1 paper)

[3] Bioinformatics (2021 – 1 paper)

[2] IEEE Transactions on Parallel and Distributed Systems (2020 – 1 paper)

[1] IEEE Transactions on Computers (2020 – 1 paper)

Reviewer Work for Technical Journals, Conferences, and Publishers pre 2020

[41] ACM Transactions on Computer Systems

[40] ACM Transactions on Programming Languages and Systems

[39] ACM Transactions on Parallel Computing

[38] ACM Transactions on Architecture and Code Optimization

[37] ACM Transactions on Embedded Computing Systems

[36] ACM Transactions on Reconfigurable Technology and Systems

[35] ACM Computer Architecture Letters

[34] IEEE Micro

[33] IEEE Computer

[32] IEEE Spectrum

[31] IEEE Transactions on Parallel and Distributed Systems

[30] IEEE Transactions on Computers

[29] IEEE Access

[28] The Computer Journal

[27] Proceedings of the VLDB Endowment

[26] Journal of Parallel and Distributed Computing

[25] International Journal of Parallel Programming

[24] IBM Journal of Research and Development

[23] BMC Bioinformatics Journal

[22] ETRI Journal

[21] IET Computers and Digital Techniques

[20] Springer Journal of Signal Processing Systems

[19] International Journal of High-Performance Computing Applications

[18] Elsevier Journal of Systems and Software

[17] Transactions on High Performance and Embedded Architectures and Compilation

[16] Journal of VLSI for Signal Processing

[15] Future Generation Computer Systems

[14] Scientific Programming

[13] EURASIP Journal on Embedded Systems

[12] Sustainable Computing: Informatics and Systems

[11] Software Practice and Experience

[10] Elsevier Journal of Network and Computer Applications

[9] Springer Nature

[8] Springer Computing

[7] Journal of Computer Science and Technology

[6] ACM Symposium on Parallel Algorithms and Architectures (SPAA)

[5] IEEE/ACM International Conference on Microarchitecture (MICRO)

[4] IEEE/ACM International Symposium on High-Performance Computer Architecture (HPCA)

[3] IEEE International Conference on Modeling and Analysis of Computer and Telecommunication Systems (MASCOTS)

[2] IEEE International Communications Conference (ICC)

[1] IEEE International Conference on Quantitative Evaluation of Systems (QEST)

Significant University and Departmental Leadership and Service

[51] Computer Science Department, Qualifying Exam Committee Chair – Systems, Networking and Cybersecurity (2026)

[50] College of Engineering Promotion and Tenure Committee, Virginia Tech (Member 2025–present)

[49] Personnel Committee, Computer Science, Virginia Tech (Chair 2025–present)

[48] Computer Science Department Head Search Committee, Virginia Tech (2023–2024)

[47] Personnel Committee, Computer Science, Virginia Tech (Associate Chair 2024–present)

[46] Personnel Committee, Computer Science, Virginia Tech (Member 2022–present)

[45] University Proposal Development Institute (PDI) Mentor, Virginia Tech (2024–present)

[44] PI Eligibility Working Group, Virginia Tech (2022)

[43] Undergraduate Program Committee, Computer Science, Virginia Tech (2021–2022)

[42] Faculty Hiring Committee, Computer Science, Virginia Tech (2021)

[41] School of Computer Science Working Group, Virginia Tech (2020, 2021)

[40] Operations Working Group, Computer Science, Virginia Tech (2020)

[39] University Reputation Group, Queen's University Belfast (2018)

[38] Global Challenges Research Fund Committee, Queen's University Belfast (2018)

[37] Director of the Queen's Global Research Institute on Electronics, Communications and Information Technology (ECIT) (2018–2019)

[36] Vice Chancellor Selection Panel, Queen's University Belfast (2017)

[35] Director of Information Services Selection Panel, Queen's University Belfast (2017)

[34] University Brand Committee, Queen's University Belfast (2017–present)

[33] Member of the University Research Forum, Queen's University Belfast (2012–2019)

[32] Chair of the University High-Performance Computing Advisory Group, Queen's University Belfast (2012–2019)

[31] Excellence in Leadership Training Program Speaker, Queen's University Belfast (2017)

[30] Member of the University Academic Council, Queen's University Belfast (2016–2018)

[29] Member of the Faculty of Engineering and Physical Sciences Promotions Panel, Queen's University Belfast (2016–2019)

[28] Member of the Faculty of Engineering and Physical Sciences Executive Board, Queen's University Belfast (2016–2019)

[27] Head of School of EEECS, Queen's University Belfast (2016–2018)

[26] Post-Doctoral Researchers Peer Mentor, Queen's University Belfast (2017–2018)

[25] Early-Career Academic Staff Peer Mentor, Queen's University Belfast (2012–2019)

[24] Senior Academic Staff Recruitment Working Group, Queen's University Belfast (2017–2019)

[23] Acting Director of Center for Data Science and Scalable Computing, Queen's University Belfast (2016–2019)

[22] Member of the Senior Management Group, Institute of Electronics, Communication and Information Technologies, Queen's University Belfast (2016–2019)

[21] Engineering (Unit of Assessment 12) REF Champion, Queen's University Belfast (2018–2019)

[20] Computer Science (UoA11) REF Champion, Queen's University Belfast (2015–2019)

[19] School of EEECS Research Strategy Group Co-Chair, Queen's University Belfast (2015–2019)

[18] BCS Program Accreditation Committee Group Member, Queen's University Belfast (2013)

[17] Computer Science Building Project Implementation Group, Queen's University Belfast (2013–2017)

[16] Excellence in Leadership Training Program, Queen's University Belfast (2012–2013)

[15] Member of the Senior Management Group, School of EEECS, Queen's University Belfast (2012–2018)

[14] School of EEECS Education Committee, Queen's University Belfast (2012)

[13] School of EEECS Academic Staff Hiring Panels, Queen's University Belfast (2012–2019)

[12] Undergraduate Curriculum Committee in Computer Science, University of Crete (2010–2012)

[11] University Data Center Infrastructure Committee, University of Crete (2010–2012)

[10] Graduate Admissions Committee in Computer Science, University of Crete (2009–2011)

[9] Computer Organization Course Coordinator, Virginia Tech (2008–2009)

[8] Junior Faculty Mentor, Virginia Tech (2008–2009)

[7] Ph.D. Qualifying Exam Committee, Virginia Tech (2007–2009)

[6] Computer Science Computing Resources Committee, Virginia Tech (2006–2008)

[5] Computer Science Graduate Admissions Committee, College of William & Mary (2005–2006)

[4] Computer Science Graduate Curriculum Committee, College of William & Mary(2005–2006)

[3] Computer Science Faculty Hiring Committee, College of William & Mary (2002–2005)

[2] Computer Science Equipment Committee, College of William & Mary (2003–2005)

[1] Freshman Academic Advisor, College of William & Mary (2004–2006)

Significant External Leadership and Service

[47] Computing Research Association (CRA), Leadership Academy Committee (2024–present)

[46] American Association for the Advancement of Sciences (AAAS), Session Reviewer (2024)

[45] External Accreditation and Evaluation Panel, Graduate Program, Master's in Data Science and Machine Learning, Master's in Information Systems, Hellenic Open University (2023)

[44] External Accreditation and Evaluation Panel, Undergraduate Program, Department of Informatics and Communications, University of Ioannina (2022)

[43] Scientific Advisory Board, Kevin Tier-2 HPC Center, Queens University Belfast (2021–present)

[42] Faculty Appointment Committee, Norwegian University of Science and Technology, Norway (2022)

[41] External Evaluation Panel, Institute of Informatics and Telecommunications, National Center for Scientific Research "Demokritos", Greece (2022)

[40] Evaluation Panel, Institute of Computer Science, Foundation for Research and Technology – Hellas, Seed Grants (2021–present)

[39] Scientific Advisory Board, Marie Curie Individual Fellowship (2022)

[38] Faculty Appointment and Promotion Committee, University of Ioannina (2022)

[37] Faculty Appointment Committee, Heidelberg University (2021)

[36] External Accreditation and Evaluation Panel, Undergraduate Program, Department of Information Systems, Ionian University (2021)

[35] External Accreditation and Evaluation Panel, Undergraduate Program, School of Electrical and Computer Engineering, National Technical University of Athens (2021)

[34] Scientific Advisory Board Member, Institute of Computer Science, Foundation for Research and Technology Hellas (FORTH), (2020–present)

[33] Faculty Appointment and Promotion Committee, National Technical University of Athens (2020)

[32] Evaluation Panel, Foundation for Research and Technology – Hellas, Synergy Grants (2019–present)

[31] Faculty Appointment and Promotion Committee, Technical University of Crete (2019)

[30] Faculty Appointment and Promotion Committee, University of Thessaly (2019, 2016)

[29] Engineering and Physical Research Council UK (EPSRC) Strategic Advisory Team (SAT) Member – e-Infrastructure (2018–present)

[28] European Commission Framework Programme Consultant on Cloud Computing, European Commission DG Unit (2018)

[27] Faculty Appointment and Promotion Committee, University of Thessaly (2018)

[26] Faculty Appointment and Promotion Committee, Université Paris-Sorbonne (2018)

[25] Knowledge Transfer Partnership in Blockchain Technology, Vox Financial Partnerships (2018)

[24] Knowledge Transfer Partnership in OpenCL Technology, Crevinn (2017)

[23] Foreign Direct Investment Consultant, Invest Northern Ireland (2017)

[22] Faculty Appointment and Promotion Committee, Aristotle University of Thessaloniki (2017)

[21] Faculty Appointment and Promotion Committee, Chalmers University of Technology (2017)

[20] Faculty Appointment and Promotion Committee, University of Crete (2020,2018,2017,2016)

[19] Scientific Advisory Board Member, European Commission Horizon2020 INTERTWinE Project (2015–2018)

[18] Scientific Advisory Member, European Commission Horizon 2020 Programme (2016)

[17] Grant Proposal Evaluator, King Abdullah University of Science and Technology (2016)

[16] Faculty Appointment and Promotion Committee, National Technical University of Athens (2016, 2014)

[15] Faculty Appointment and Promotion Committee, Ionian University (2016)

[14] Faculty Appointment and Promotion Committee, University of Athens (2015)

[13] Faculty Appointment and Promotion Committee, Technological Educational Institute of Athens (2015)

[12] Faculty Appointment and Promotion Committee, Technological Educational Institute of Piraeus (2015)

[11] Faculty Appointment and Promotion Committee, Technological Educational Institute of Western Greece (2015,2016)

[10] Faculty Appointment and Promotion Committee, Technological Educational Institute of Piraeus (2014)

[9] Faculty Appointment and Promotion Committee, Technological Educational Institute of Western Greece (2014)

[8] Industry-Academic Partnership Training, Westminster Higher Education Forum (2013)

[7] External examiner, School of Computing, University of Leeds (2012–present)

[6] Faculty Appointment and Promotion Committee, University of Crete (2013)

[5] Multiple Industrial Consultancies, Queen's University Belfast (2012–present)

[4] Faculty Appointment and Promotion Committee, University of Ioannina (2011)

[3] Faculty Appointment and Promotion Committee, Aristotle University of Thessaloniki (2011)

[2] Faculty Hiring Committee, Technical University of Denmark (2010)

[1] Faculty Hiring Committee, National Technical University of Athens (2010)

Ph.D. Examiner (UK) or Committee Member (US and elsewhere)

[55] **Ritvik Prabhu**. Department of Computer Science, Virginia Tech, 2026–present.

[54] **Sijia Li**. Department of Computer Science, Virginia Tech, 2026–present.

[53] **Hanry Xu**. Department of Computer Science, Virginia Tech, 2026–present.

[52] **Hamid Hadian**. Department of Computer Science, Virginia Tech, 2023–present.

[51] **Hafiz Adnan Niaz**. Department of Computer Science, University College Dublin, 2025.

[50] **Sumit Monga**. Department of Electrical and Computer Engineering, Virginia Tech, 2025.*Programming Abstractions for Scalable and High-Performance Memory Disaggregation*

[49] Hayden Estes. Department of Computer Science, Virginia Tech, 2025.

[48] **Shoaib Asif Qazi**. Department of Computer Science, Virginia Tech, 2025.

[47] **Frank Wayne**. Department of Computer Science, Virginia Tech, 2025. *Fast and Accurate Graph Clustering*.

[46] **Haihang Wu**. Department of Mechanical Engineering, University of Melbourne, 2025. *Towards Efficient and Robust Deep Neural Networks*.

[45] **Jing Chen**. Department of Computer Science and Engineering, Chalmers University of Technology, 2024. *Adaptive Task Scheduling and Resource Management Techniques for Energy Efficiency on Multi-core Architectures*.

[44] **Madhava Krishnan**. Department of Electrical and Computer Engineering, Virginia Tech, 2021 *Designing Scalable System Software for Emerging Storage Technologies*. Supervisor: Changwoo Min.

[43] **Anastasios Papagiannis**. Department of Computer Science, University of Crete, 2020 *Memory Mapped I/O For Fast Storage*. Supervisor: Angelos Bilas.

[42] **Simar Preet Singh**. Department of Computer Science & Engineering, Thapar University, 2020 *Energy-Efficient Load Balancing Algorithms in Fog Computing*. Supervisor: Rajesh Kumar.

[41] **Song Zheng**. Department of Computer Science, Virginia Tech, 2020 *Self-Adaptive Edge Services: Enhancing Reliability, Efficiency, and Adaptiveness under Unreliable, Scarce, and Dissimilar Resources*. Supervisor: Eli Tilevich.

[40] **Dimitrios Chasapis**. Barcelona Supercomputing Center, 2019 *Towards Resource-Aware Computing for Task-Based Runtimes and Parallel Architectures*. Supervisor: Marc Casas Guix, Miquel Moretó Planas.

[39] **Leszek Sliwko**. University of Westminster, 2019 *Intelligent Load Balancing in Cloud Computer Systems*. Supervisor: Vladimir Getov.

[38] **Mohammed Al-Hayanni**. Newcastle University, 2018 *Investigation into Scalable Energy and Performance Models for Many-Core Systems*. Supervisor: Alex Yakovlev.

[37] **Daniele di Sensi**. University of Pisa, 2018 *Self-Adaptive Solutions for Managing Performance and Power Consumption of Parallel Applications*. Supervisor: Marco Danelutto.

[36] **Ahsan Javed Awan**. KTH, 2017 *Performance Characterization and Optimization of In-Memory Data Analytics on a Scale-up Server*. Supervisor: Mats Brorsson.

[35] **Vassilis Vassiliadis**. University of Thessaly, 2017 *Optimization of Program Execution using Computational Significance*. Supervisor: Christos D. Antonopoulos.

[34] **Rajiv Nishtala**. Barcelona Supercomputing Center, 2017 *Energy Optimising Methodologies on Heterogeneous Data Centers*. Supervisor: Xavier Martorell.

[33] **Foivos Zakkak**. Computer Science, University of Crete, 2016 *Java on Scalable Memory Architectures*. Supervisor: Polyvios Pratikakis.

[32] **Ioannis Nikolakopoulos**. Computer Science and Engineering, Chalmers University, 2016 *Shared Memory Objects as Synchronization Abstractions: Algorithmic Implementations and Concurrent Applications*. Supervisor: Marina Papatriantafillou.

[31] **Spiros Agathos**. Computer Engineering, University of Ioannina, 2016 *Efficient OpenMP Runtime Support for General-Purpose and Embedded Multi-core Platforms*. Supervisor: Vassilis Dimakopoulos.

[30] **Madhavan Manivannan**. Computer Science and Engineering, Chalmers University, 2016 *Towards Runtime-Assisted Cache Management for Task-Parallel Programs*. Supervisor: Per Stenström.

[29] **Kiran Chandramohan**. Informatics, University of Edinburgh, 2016 *Mapping Parallelism to Heterogeneous Processors*. Supervisor: Michael O'Boyle.

[28] **Javier Bueno Hedo**. Computer Architecture, Universitat Politecnica de Catalunya, 2015 *Runtime Support for Multi-Level Disjoint Memory Address Spaces*. Supervisor: Xavier Martorell.

[27] **Eleftherios Kosmas**. Computer Science, University of Crete, December 2014. *Techniques for Enhancing Parallelism in Mechanisms that Automatically Execute Sequential Code in Concurrent Environments*. Supervisor: Panagiota Fatourou.

[26] **Georgios Vassiliadis**, Computer Science, University of Crete, December 2014. Thesis title: *Accelerating Stateful Network Packet Processing Using Graphics Hardware*. Supervisor: Evangelos Markatos, Sotiris Ioannidis.

[25] **Chun-Yi Su**, Computer Science, Virginia Tech, December 2014. Thesis title: *Resource Management on Heterogeneous Multi-Core, Multi-Memory Systems*. Supervisor: Kirk W. Cameron.

[24] **Hung-Ching Chang**, Computer Science, Virginia Tech, December 2014. Thesis title: *Measuring, modeling, and optimizing counterintuitive performance phenomena in power-scalable, parallel systems*. Supervisor: Kirk W. Cameron.

[23] **Muhammad Tayyab Chaudhry**, Computer Science and Information Technology, University of Malaya, December 2014. Thesis title: *Thermal-Aware Scheduling in Green Data Centers*. Supervisor: Ling Teck Chaw.

[22] **Pranav Tendulkar**, Computer Science, Verimag and University of Grenoble, France, October 2014. Thesis title: *Mapping and Scheduling on Multicore Processors using SMT Solvers*. Supervisor: Oded Maler.

[21] **Iasonas Polakis**, Computer Science, University of Crete, Greece, February 2014. Thesis title: *Online Social Networks from a Malicious Perspective: Novel Attack Techniques and Defense Mechanisms*. Supervisor: Evangelos Markatos.

[20] **Anastasios Nanos**, Electrical and Computer Engineering, National Technical University of Athens, Greece, December 2013. Thesis title: *Efficient I/O Resource Sharing in Virtual Machine Environments*. Supervisor: Nectarios Koziris.

[19] **Nikolaos Kallimanis**, Computer Science, University of Ioannina, Greece, May 2013. Thesis title: *Highly Efficient Synchronization Techniques in Shared Memory Distributed Systems*. Supervisor: Panagiota Fatourou.

[18] **Mushen Owaida**, Computer & Communication Engineering, University of Thessaly, Greece, September 2012. Thesis title: *Using Parallel Programming Models for Architectural Synthesis*. Supervisor: Nikolaos Bellas.

[17] **Carlos Villavieja**, Computer Architecture, Universitat Politecnica de Catalunya, January 2012. Thesis title: *Hardware and Software Support for Distributed Shared Memory in Chip Multiprocessors*. Supervisor: Alex Ramirez.

[16] **Demetrios Antoniades**, Computer Science, University of Crete, December 2011. Thesis title: *Understanding File and Information Sharing Services in Web 2.0*. Supervisor: Evangelos Markatos.

[15] **Mauricio Alvarez**, Computer Architecture, Universitat Politecnica de Catalunya, September 2011. Thesis title: *Parallel Video Decoding*. Supervisor: Alex Ramirez.

[14] **Elias Athanasopoulos**, Computer Science, University of Crete, March 2011. Thesis title: *Modern Techniques for the Detection and Prevention of Web 2.0 Attacks*. Supervisor: Evangelos Markatos.

[13] **Andrea Di Biaggio**, Electronics and Informatics, Politecnico di Milano, December 2010. Thesis title: *Synchronization and Data Distribution Optimization for Distributed Shared Memory Multiprocessors*. Supervisor: Stefano Crespi Reghizzi.

[12] **Stamatis Kavadias**, Computer Science, University of Crete, September 2010. Thesis title: *Direct Communication and Synchronization Mechanisms in Chip Multiprocessors*. Supervisor: Manolis Katevenis.

[11] **Kornilios Kourtis**, Electrical and Computer Engineering, National Technical University of Athens, April 2010. Thesis title: *Data Compression Techniques for Performance Improvement of Memory-Intensive Applications on Shared Memory Architectures*. Supervisor: Nectarios Koziris.

[10] **Nikolaos Anastopoulos**, Electrical and Computer Engineering, National Technical University of Athens, March 2010. Thesis title: *Techniques for the Optimization and Efficient Mapping of Parallel Code on Computational Nodes with Multithreaded and Multicore Processors*. Supervisor: Nectarios Koziris.

[9] **Dimitrios Syrivelis**, Computer & Communication Engineering, University of Thessaly, June 2009. Thesis title: *Exploiting Reconfigurable Heterogeneous Parallel Architectures in a Multitasking Context: a Systems Approach*. Supervisor: Spyros Lalis.

[8] **Matthew Tolentino**, Computer Science, Virginia Tech, February 2009. Thesis title: *Managing Memory for Power, Performance, and Thermal Efficiency*. Supervisor: Kirk W. Cameron.

[7] **Guanying Wang**, Computer Science, Virginia Tech, September 2009. Thesis title: Evaluating MapReduce Systems: A Simulation Approach. Supervisor: Ali R. Butt

[6] **Montse Farreras**, Computer Architecture, Universitat Politecnica de Catalunya, December 2008. Thesis title: *Optimizing Programming Models for Massively Parallel Computers*. Principal Supervisor: Toni Cortes.

[5] **Andrey Chernikov**, Computer Science, College of William & Mary, August 2007. Thesis title: *Parallel Generalized Delaunay Mesh Refinement*. Supervisor: Nikos Chrisochoides.

[4] **Qi Zhang**, Computer Science, College of William & Mary, December 2006. Thesis title: *The Effect of Workload Dependence in Systems: Experimental Evaluation, Analytic Models, and Policy Development*. Supervisor: Evgenia Smirni.

[3] **Songqing Chen**, Computer Science, College of William & Mary, August 2004. Thesis title: *Building Internet Caching Systems for Multimedia Content Delivery*. Supervisor: Xiaodong Zhang.

[2] **Kevin Barker**, Computer Science, College of William & Mary, May 2004. Thesis title: *Runtime Support for Load Balancing of Parallel Adaptive and Irregular Applications*. Supervisor: Nikos Chrisochoides.

[1] **Zhichun Zhu**, Computer Science, College of William & Mary, August 2003. Thesis title: *Power Considerations for Memory-related Microarchitecture Designs*. Supervisor: Xiaodong Zhang.

Government Research Funding Panelist & Reviewer Service

[42] National Science Foundation (NSF). Panelist, 2025 (two programs)

[41] Swiss National Science Foundation. Grant Proposal Reviewer, 2025, 2020, 2016

[40] French National Research Agency (ANR). Panelist, 2025

[39] Hellenic Foundation for Research and Innovation. Greece. Panelist, 2025

[38] National Science Foundation. Computer and Information Science and Engineering. US. Panelist, 2024

[37] Hellenic Foundation for Research and Innovation. Greece. Panelist, 2024

[36] National Science Foundation. Computer and Information Science and Engineering. US. Panelist, 2023

[35] National Science Foundation. Computer and Information Science and Engineering. US. Panelist, 2021

[34] Coastal Virginia Center for Cyber Innovation (COVA CCI). US. Grant Proposal Reviewer, 2021

[33] Knowledge Foundation Sweden. Grant Proposal Reviewer, 2021

[32] Italian National Research Council (CINECA). Grant Proposal Reviewer, 2021

[31] European Research Council (ERC). Grant Proposal Reviewer, 2020

[30] US Department of Energy. Grant Proposal Reviewer, 2020

[29] Royal Academy of Engineering, United Kingdom. Grant Proposal Reviewer, 2017

[28] Grant Proposal Evaluator, Austrian Academy of Sciences, 2017

[27] Technology Foundation STW, The Netherlands. Grant Proposal Reviewer, 2016

[26] National Science Center, Poland. Grant Proposal Reviewer, 2016

[25] Natural Science and Engineering Research Council of Canada (NSERC). Discovery Grants Panelist, 2016

[24] University of Cyprus Research Foundation. Grant Proposal Reviewer, 2016

[23] Royal Academy of Engineering, United Kingdom. Grant Proposal Reviewer, 2015

[22] Natural Science and Engineering Research Council of Canada (NSERC). Discovery Grants Panelist, 2015

[21] Natural Science and Engineering Research Council of Canada (NSERC). Discovery Grants Panelist, 2014

[20] UK Engineering and Physical Sciences Research Council (EPSRC). Grant Proposal Reviewer, 2021

[19] UK Engineering and Physical Sciences Research Council (EPSRC). Platform Grant Panelist, 2015

[18] UK Engineering and Physical Sciences Research Council (EPSRC). Grant Proposal Reviewer, 2018

[17] UK Engineering and Physical Sciences Research Council (EPSRC). Grant Proposal Reviewer, 2017

[16] UK Engineering and Physical Sciences Research Council (EPSRC). Grant Proposal Reviewer, 2014

[15] UK Engineering and Physical Sciences Research Council (EPSRC). Grant Proposal Reviewer, 2013

[14] UK Engineering and Physical Sciences Research Council (EPSRC). Grant Proposal Reviewer, 2012

[13] European Commission FP7 Framework Programme. Project Reviewer, 2016

[12] European Commission FP7 Framework Programme. Project Reviewer, 2015

[11] European Commission FP7 Framework Programme. Project Reviewer, 2014

[10] European Commission FP7 Framework Programme. Project Reviewer, 2013

[9] European Commission FP7 Framework Programme. Grant Proposal Reviewer, 2012

[8] Greek Secretariat for Research and Technology. Grant Proposal Reviewer, 2010

[7] U.S.–Israel Binational Science Foundation. Grant Proposal Reviewer, 2009

[6] United States National Science Foundation. Panelist, 2008

[5] Natural Science and Engineering Research Council of Canada. Grant Proposal Reviewer, 2007

[4] United States National Science Foundation. Panelist, 2004

[3] United States National Science Foundation. Panelist, 2003

[2] United States National Science Foundation. Panelist, 2002

[1] Maryland Industrial Partnerships Program. Grant Proposal Reviewer, 2007

Conference Panels

[8] *Fifth International Workshop on Extreme Scale Programming Models and Middleware*. Held in conjunction with SC2020. When one size doesn't fit all: programming models in the post-Moore era. Panelist. November 2020.

[7] *EPSRC Workshop on Manycore Computing: Hardware and Software*. Panelist. Southampton, UK, January 2018.

[6] Heterogeneous and/or Homogeneous computing supporting parallel applications Which are the key driving factors for the application developers and platform designers? Are they cooperating or fighting? *6th Workshop on Parallel Programming and Run-Time Management Techniques for Many-core Architectures* ***(PARMA-DITAM 2015)*** . Panelist. January 2015.

[5] *IBM Research ExaChallenge Symposium*. Panelist. Dublin, Ireland, October 2012.

[4] Accelerators: Fad, Fashion, or Future? *39th International Conference on Parallel Processing* ***(ICPP)***. Panelist. September 2010.

[3] Key Challenges Presented by Next Generation Hardware Systems. *Fall Creek Falls Conference*. Panelist. September 2007.

[2] Invitee, *Microsoft Faculty Research Summit*. Redmond, WA, September 2007.

[1] NSF Next Generation Systems Software Program. *15th ACM International Conference on Supercomputing* ***ICS***. Panelist. June 2001.

Public Engagement

[12] More to Know Podcast with Christopher Conover. The Future of Datacenters: AI, Energy Demand, and the Next Infrastructure Boom. link (also on Apple and Spotify). February 2026.

[11] TedX MidAtlantic Talk on Myths and Ways Forward for AI. link. November 2025.

[10] Northwest Indiana Times. Data Centers in Northwest Indiana: Key Issues and Concerns. link. December 2025.

[9] Fuel Cell Energy. On the Societal and Economic Implications of Datacenters. link. December 2025.

[8] MSN and WFXR. On the Societal and Economic Implications of Datacenters. link 1, link 2. December 2025.

[7] Cardinal News. On the Societal and Economic Implications of Datacenters. link May 2025.

[6] VT Science Corner. A Way to Democratize AI. Roanoke Times. link. January 2024.

[5] Queen's Researchers in Bid to Develop the Fastest Supercomputers. Newswise. link. February 2015.

[4] NNTV Behind the Science. On the Importance of Supercomputers. 2015.

[3] Horizon2020 Focus Workshops in Northern Ireland and the UK. March–September 2014.

[2] Faster than 50 Million Laptops: The Race Goes to Exascale. Interview link

[1] Big Iron Move Toward Exascale. Interview link

Invited Seminars and Talks

[55] *SWEET: A Hardware & Software Stack for Wrist-to-Cloud Intelligence*. NSF CSR Principal Investigators Meeting, Boca Raton, FL, September 2025.

[54] *SWEET: System Architectures and Programming Abstractions for Resilient and Efficient Edge AI*. Discover-US Webinar, European Commission, Virtual Event, April 2025.

[53] *Sustainable AI from the Viewpoint of an AI Skeptic*. Distinguished Speaker Seminar, Department of Computer Engineering and Informatics, Virtual Event, December 2024.

[52] *Programming Models for High-Performance Computing: Key Drivers and Barriers to Adoption*. Invited Talk, First NI-HPC Users Conference, Virtual Event, October 2021.

[51] *Why Should We Still be Concerned about Energy Management in System Software?* Invited Talk, 12th International Green and Sustainable Computing Conference, Virtual Event, October 2021.

[50] *From Approximate to Significance-Driven to Transprecision Computing: Challenges and Opportunities*. Distinguished Speaker Seminar, Department of Computational Science, ETH Zurich, April 2018.

[49] *Virtualizing Any Accelerator Anywhere*. Distinguished Speaker Seminar, Department of Computer Science, College of William and Mary, January 2018.

[48] *The Jevons Paradox in Computing Systems Research*. Distinguished Lecture Series, Department of Computer Science, Virginia Tech, November 2016.

[47] *Computational Significance and its Implications for HPC*, 13th Workshop on Clusters, Clouds, and Data for Scientific Computing **(CCDSC'16)** , Chemin de Chanzé, France, October 2016.

[46] *Computational Significance and its Implications for Computing Systems*, School of Electrical and Electronic Engineering, Newcastle University, October 2016.

[45] *Scaling Up, Out, or Down*, School of Informatics, University of Edinburgh, March 2016.

[44] *Significance-Driven Runtime Systems*, RoMoL'16 Workshop, Barcelona, Spain, March 2016.

[43] *Advances in Energy-Efficient and Resilient HPC: Scaling Up, Out, or Back?*, Cardiff University, March 2016.

[42] *Variability: Why should we care?*, Birds of a Feather Session on Variability in Large-Scale Computing Systems, held in conjunction with the SC'15 Conference, Austin, TX, November 2015.

[41] *New Approaches to Energy-Efficient and Resilient HPC*, Department of Computer Science, Old Dominion University, November 2015.

[40] *HPDC Research at Queen's: An Overview*, ARM High-Performance Computing Group, Manchester, UK, November 2015.

[39] *Server Resource Provisioning for Real-Time Analytics using Iso-Metrics*, Workshop on Performance Modeling: Methods and Applications, in conjunction with the 2015 International Supercomputing Conference **(ISC'15)** , Frankfurt, Germany, July 2015.

[38] *Evaluating Servers using Iso-Metrics: Power, Performance and Programmability Implications.* Eighth Workshop on Programmability Issues for Heterogeneous Multicores **(MULTIPROG'15)**, Amsterdam, The Netherlands, January 2015.

[37] *The Challenges and Opportunities of Micro-Servers in the HPC Ecosystem*, 12th Workshop on Clusters, Clouds, and Data for Scientific Computing **(CCDSC'14)** , Chemin de Chanzé, France, October 2014.

[36] *NVRAM as a User-Level Object Store.* HiPEAC Autumn Computing Systems Week, Athens, Greece, October 2014.

[35] *On the Viability of Microservers for Real-Time Data Analytics.* HiPEAC Autumn Computing Systems Week, Athens, Greece, October 2014.

[34] *NanoStreams: A Hardware and Software Stack for Real-Time Analytics on Fast Data Streams*. Horizon 2020 – the HPC Opportunity, London, United Kingdom, March 2014.

[33] *GEMSCLAIM: Greener Mobile Systems by Cross-Layer Energy Management*. CHIST-ERA 2014 Projects Seminar, Istanbul, Turkey, March 2014.

[32] *Searching for Data: The Ever-Increasing Role of Memory Hierarchies on the Performance and Sustainability of Computing Systems*. Inaugural Lecture, Queen's University of Belfast, March 2013.

[31] *Energy as a Resource in Parallel Programs*. Supercomputing'12 Birds-of-a-Feather Session on *Cool Supercomputing*, November 2012.

[30] *Block-Level Dynamic Dependence Analysis for Task-Based Parallelism*. *Workshop on Perspectives on Parallel Numerical Linear Algebra*, Manchester, UK, July 2012.

[29] *Software Techniques for Energy Conservation in High-End Computing Systems*. Invited Seminar, School of Computer Science, University of Manchester, UK, March 2012.

[28] *Energy Efficiency at Extreme Scale Tools and Challenges*. Supercomputing'11 Birds-of-a-Feather Session on *Energy-Efficiency*, November 2011.

[27] *Rearchitecting MapReduce for Heterogeneous Multicore Processors with Explicitly Managed Memories*. School of Electronics, Electrical Engineering and Computer Science, Queen's University of Belfast. October 2010.

[26] *Determinism in Parallel Software and Architectures*. HiPEAC Systems Week Cluster Meetings, Barcelona, October 2009.

[25] *Parallelizing Non-trivial Applications with Multiple Programming Models*. HiPEAC Systems Week Cluster Meetings, Barcelona, October 2009.

[24] *Uniform Evaluation of Programming Models*. HiPEAC Systems Week Cluster Meetings, Paris, November 2008.

[23] *Unifying Layered Parallelism on the Cell BE*. Supercomputing'07 Birds-of-a-Feather Session on *Unleashing the Power of the Cell Broadband Engine Processor for HPC*, November 2007.

[22] *Unified Scheduling of Polymorphic Parallelism on Asymmetric Multi-core Systems*. Lawrence Livermore National Laboratory. October 2007.

[21] *System Software for Scaling on Many Cores*. Oak Ridge National Laboratory. September 2007.

[20] *Design and Implementation of Time- and Power-Efficient Software Stacks for Multicore Processors*. IBM Thomas J. Watson Research Center. December 2006.

[19] *Design and Implementation of Time- and Power-Efficient Software Stacks for Multicore Processors*. Department of Computer Science, North Carolina State University. September 2006.

[18] *Hardware Event-Driven Scalability Predictors: Improving Energy-Efficiency under Hard Performance Constraints on Multi-core and Multi-threaded Architectures*. Department of Electronic and Computer Engineering, Technical University of Crete. June 2006.

[17] *Addressing the Challenges of Chip Multiprocessors using Autonomic Software*. Department of Computer Science, University of California, Riverside. April 2006.

[16] *Addressing the Challenges of Chip Multiprocessors using Autonomic Software*. Department of Electrical and Computer Engineering, University of British Columbia. March 2006.

[15] *High-Performance Power-Efficient Runtime Environments for Dense Computing Systems*. Department of Computer Science, Virginia Tech. February 2006.

[14] *High-Performance Power-Efficient Runtime Environments for Dense Computing Systems*. Institute of Computer Science, Foundation for Research and Technology – Hellas. June 2005.

[13] *High-Performance Power-Efficient Runtime Environments for Dense Computing Systems*. Department of Computer Engineering and Informatics, University of Patras. June 2005.

[12] *A Unified Programming Framework for Multigrain Multithreaded Architectures*. Institute of Computer Science, Foundation for Research and Technology – Hellas. June 2004.

[11] *A Unified Programming Framework for Multigrain Multithreading*. School of Electrical and Computer Engineering, National Technical University of Athens. June 2004.

[10] *A Unified Programming Framework for Multigrain Multithreading*. Department of Computer Science, University of California at Riverside. April 2004.

[9] *A Unified Programming Framework for Multigrain Parallel Architectures*. Department of Electrical and Computer Engineering, Northwestern University. February 2004.

[8] *Program Transformations and Scheduling Algorithms for Managing Shared Caches on Multithreaded Processors*. Department of Informatics, Athens University of Economics and Business. June 2003.

[7] *Program Transformations and Scheduling Algorithms for Managing Shared Caches on SMT Processors*. IBM Thomas J. Watson Research Center. March 2003.

[6] *Building Adaptive Programs with Local Sensing of Execution Conditions*. Department of Computer Science, Texas A&M University. March 2003.

[5] *Interoperable System Software*. Department of Information and Computer Sciences, University of California, Irvine. April 2002.

[4] *Interoperable System Software*. Department of Computer Science, College of William and Mary. March 2002.

[3] *Scaling Shared-Memory Programming Models beyond Shared-Memory Architectures*. Department of Computer Science, University of Houston. November 2001.

[2] *Some Steps towards Simple, Scalable and Portable Parallel Programming Models*. Department of Computer Science, College of William & Mary. October 2001.

[1] *A Case for User-Level Page Migration*. Coordinated Sciences Laboratory, University of Illinois at Urbana-Champaign. January 2001.